

Review

Comprehensive analysis of IPMSM drives with FOC under different operating modes

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Abstract

Electrical drives play an essential role in electromechanical energy conversion. Drive systems have a variety of applications, including home appliances and industries such as manufacturing processes, transportation, and wind energy fields. The main technology of a high-performance robust drive system is a high-efficiency motor and proper sensorless observer. Permanent magnet synchronous motors (PMSMs), according to their relative advantages, are commonly used for high-performance traction and robotic systems. PMSMs have the merits of a high power-to-weight ratio, high torque per ampere ratio, and high efficiency. PMSMs may be surface mounted (SM-PMSM) or interior type (IPMSM) according to the rotor design. IPMSMs are commonly used for traction systems due to their merits over SM-PMSMs, such as higher torque, robust design, higher efficiency, and a wider range of constant-power regions. The IPMSM is operated under the maximum torque per ampere (MTPA) operating mode for low and medium speeds and under the field weakening (FW) operating mode for the constant-power region. This paper reviews two drives based on field-oriented control (FOC) and finite control set-model predictive control (FCS-MPC) techniques for an IPMSM drive system and proposes and implements the IPMSM drive system based on FOC with a conventional PI controller under different operating modes.

Keywords IPMSM · Field-oriented control · MPC · Speed sensorless · Control · 2L-VSI · MCs · PWM

1 Introduction

Electrical drives are valuable for their role in electromechanical energy conversions. Drive systems are used everywhere in daily life, home appliances and industries such as cooling fans, washing machines, manufacturing processes, transportation, mineral processing, wind energy fields, computer peripherals and many other industry applications. Via a magnetic field, any drive system can control the conversion of electrical energy into mechanical energy. The electrical machine that has the main role in this conversion is the workhorse in a drive system. Almost all the electrical energy generated around the world is consumed by electric motors [1]. Thus, if electric motors operate efficiently by special design and proper control of these machines, then the operation will save considerable energy worldwide [2]. Electric vehicles are widely applicable electrical drives that are environmentally friendly (no harmful local gas emissions, minimum global emissions, or low noise) and unbinding for city traffic problems [3]. Currently, EVs can replace traditional vehicles, which are based on internal combustion engines, facing the soaring prices of fossil fuel, reducing carbon emissions, and becoming an environmentally friendly technology [4, 5].

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The main technology for developing a high-performance drive system is suitable electric machine selection and proper control systems because the machine type and its characteristics are essential factors for drive system model success [5]. The proper motor is selected based on its power density, efficiency, controllability, reliability, cost, noise, and maintenance [6, 7].

PMSMs have more advantages than other drivers, including no rotor copper loss, an effective power factor, high efficiency, a high torque-to-inertia ratio, a high power-to-weight ratio, and robustness. However, PMSMs suffer from [6–8]:-

- High permanent magnet (PM) (Neodymium iron boron (Nd-Fe-B)) material costs, because of its high energy density;
- Uncontrollable rotor PM flux;
- Possible demagnetization effects, due to armature reaction effect, transient and machine fault conditions;
- Unlike the conventional three-phase synchronous motor, PMSMs suffer from lack of power factor (reactive power) control and its association with terminal voltage regulation.

IPMSMs are superior among PMSMs due to their advantages, which include the saliency design of the machine, faster dynamic response, rugged and robust construction because the PMs are embedded into the rotor, and higher efficiency over a wider speed range for the field weakening technique [9]. However, the IPMSM armature reaction effect is dominant because of the low effective air gap [8]. Because of their merits, IPMSMs are more suitable for drive systems.

In recent years, the rapid development of microcontrollers, such as digital signal processors (DSPs), and advances in control research have produced various reliable and robust control techniques for power electronics, electrical drive systems and traction applications, such as EVs [10]; however, field-oriented control (FOC) based on a conventional PI controller has been applied because of its advantages and good performance during the steady state.

In recent decades, various sensorless speed/position estimation schemes have been proposed to exclude speed sensors without affecting drive system performance to provide exact rotor speed and position. Estimating and controlling the speed is essential for high-end applications (washing machines and air conditioners) [11] and for applications where safety requirements are functional, such as EV systems [12]. Speed estimation has many advantages, such as eliminating the speed sensor and its cable, less maintenance and greater robustness, and reduced cost [13, 14].

Electric drive system research has covered a wide area in recent decades toward achieving high-performance and reliable systems for various drive systems, such as industrial, traction, and robotic systems. A high efficiency, robust driving motor, and proper sensorless control are essential factors for achieving the demand for reliable, high-performance drive systems with a wide range of constant power regions [15, 16]. Therefore, an IPMSM is used in this paper because of its relative benefits, including high efficiency, high torque with reluctance torque production, high energy-to-weight ratio, and robust design suitable for a wide range of constant power regions [17]. During low-speed operation, achieving a higher efficiency of IPMSM is a special target, while during high-speed operation, the target attains a wider range of speeds.

To achieve a high-performance drive system, providing accurate rotor speed and position is necessary. Various schemes for estimating the sensorless speed/position are proposed for eliminating the need for speed sensors without affecting the drive system performance to provide the exact rotor speed and position. A proper sensorless observer and improved control technique compared to conventional FOC based on a PI controller due to its hardness of tuning and operating constraints have been the main targets of recent decades.

The remainder of the paper is organized as follows: Sect. 2 provides a general background review of PMSMs' categories, advantages, and control techniques. This section aims to understand the research challenges facing PMSMs and their control and to identify points of research. After this section, the work is decided to be on an IPMSM and so on with the MTPA to reduce the loss and FW to obtain a higher speed suitable for electric drive systems such as electric vehicle traction applications. Section 3 illustrates the mathematical model of an IPMSM as per the space vector representation, the motor operating modes of maximum torque per ampere and field weakening, the space vector voltage source inverter model, and the space vector pulse width modulation (SV-PWM). Section 4 proposes and implements FOC for an IPMSM drive system based on a MATLAB/Simulink environment. The implementation focuses on machine operation under different operating modes (MTPA/FW) and under different operating conditions, such as changeable speeds under a constant load and sudden rated load imposition for fixed-speed operation. The simulation results validate the effectiveness of FOC based on the conventional PI controller with performance discussion.

2 Permanent magnet synchronous motor review

The main technology for developing a high-performance drive system is electric machine selection and proper sensorless control systems because the machine type and its characteristics are the main factors for successful operation of the drive system model [5]. Then, the commonly used electric motors for various drive systems, such as traction systems, EVs, and hybrid EVs (HEVs), are illustrated and compared.

2.1 Electrical machine types

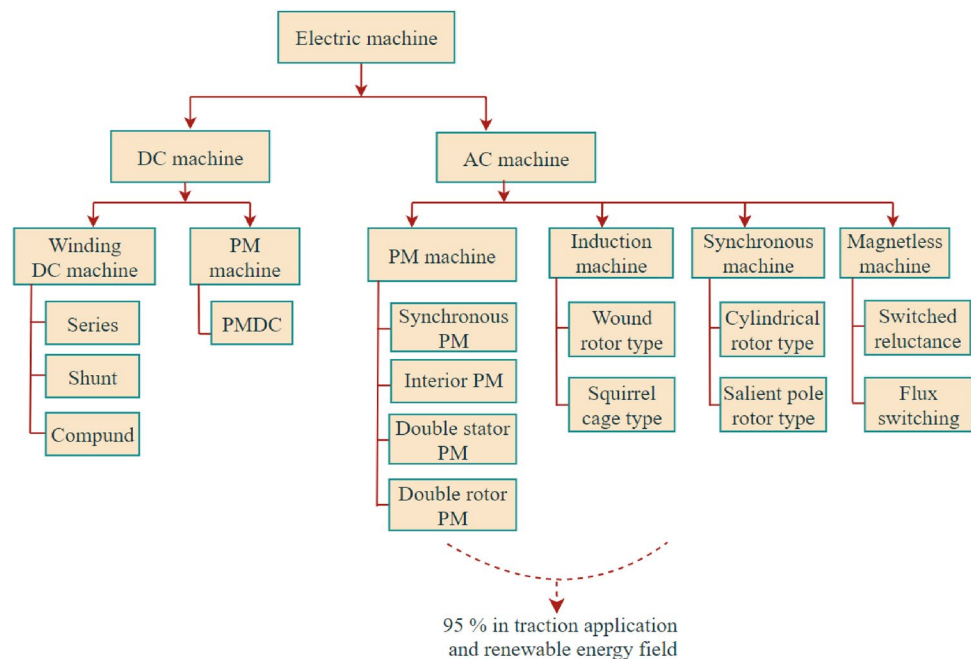
The general machine map shown in Fig. 1 classifies electrical machines into DC and AC machines, which may be brushless or brushed machines.

The essential features of an electric drive motor include high torque and power density, high starting torque, wide speed range over the constant-power region, high efficiency over wide ranges of operating conditions, high reliability and robustness, low weight, small volume, acceptable low cost, good controllability, and low torque ripple [3, 6, 15]. The most popular (nearly 95%) electric machines used for traction and renewable energy fields are induction machines and permanent magnet machines [5].

2.2 Electric drives

In 1987, R. Krishnan [18] reviewed the AC servo motor drive family, which consists of PMSM, DC brushless, induction, and switched reluctance machines (SRMs). The authors stated that ac motor drives for servo applications are more attractive than dc brushed machines are. The SRMs can deliver performances for servo drives equivalent to those of DC brushed machines. PM machines may be axial or radial flux machines. PM axial machines have a higher power density and a greater torque-to-inertia ratio than radial machines with more complex manufacturing and more s. The selection of a suitable servo motor for applications is a serious problem for users and application engineers, as the potentials and pitfalls of AC servo motors are reviewed for help, such as cost, power density, acceleration rates, peak torque capability, torque per unit ampere, speed range, parameter sensitivity, ripple torque, rotor position feedback device, braking, thermal capability, and cogging torque.

Fig. 1 General Machine Map



2.2.1 DC machines

DC variable-speed drives are the most common drive systems used in industrial and EV applications [3] due to their merits of maturity and simplicity because of their simple independent control of the air gap flux and armature current. Thus, the torque is proportional to the two independent control terms of the flux and armature current. However, DC machines are not attractive for modern drives due to their low efficiency, high cost, high rotor inertia, and frequent maintenance due to their use of commutators (which results in torque ripples and limits motor speed and peak current) and brushes (which result in friction as well as radio frequency interference) [3, 19].

2.2.2 Induction machines

Induction machines are the major equipment among various commutatorless machines, especially squirrel cage motors, due to their merits, such as being commutatorless, brushless, low maintenance, low cost, and rugged.

AC machines' speed control is more complex than that of DC machines because of the nonlinear coupled model. Based on special control techniques such as variable voltage variable frequency (VVVF) control and field-oriented control (FOC), induction machines have been considered for recent drive systems such as EVs and HEVs. The VVVF technique is not adequate for high-performance drive systems because of its air-gap flux drift and slow response. FOC enables AC machines to be controlled like a separately excited DC machine [3].

2.2.3 Permanent magnet brushless machines

Permanent magnet (PM) brushless machines (especially the synchronous type) are becoming increasingly attractive due to their advantages, which are summarized as [3]:

1. High power density (reduced weight-to-power ratio) is achieved by using rotor high-energy permanent magnets, such as rare-earth permanent magnets, inserted in rotor structures, such as Neodymium-iron-boron (NdFeB) alloys;
2. No increase in the efficiency of rotor copper loss;
3. Increased rotor acceleration occurs because of the low rotor electromechanical time constant.

However, PM motors suffer from drawbacks such as the following:

1. High PM material cost;
2. Uncontrollable PM flux;
3. Possible demagnetization effects.

2.2.4 2.2.4. Switched reluctance machine

Switched reluctance machines are valuable for electric propulsion (for both EVs and HEVs) because of their simple and rugged construction, low cost, extremely high-speed operation permission, and hazard-free operation. Because of the disadvantages of low power density, control nonlinearity, inherent torque ripples, and acoustic noise, their applications for drive systems, especially traction systems (e.g., EVs and HEVs), have been limited for several years [3, 20].

In brief, the merits and demerits of the most applicable electric motors for drive systems are summarized in Table 1.

2.3 Previous work on permanent magnet synchronous motor

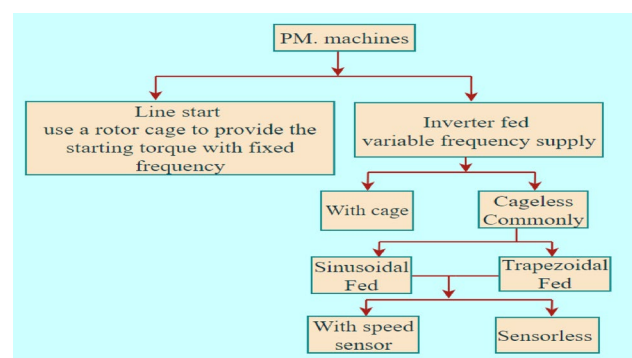
The electric drive systems field has covered a wide range of research areas in recent decades toward achieving high-performance and reliable electric drive systems. High efficiency, a robust driving motor, and proper sensorless control are essential factors in achieving a reliable, high-performance drive system [15, 16]. PMSMs provide high power-to-weight, high torque per current ratio, and high efficiency because there is no rotor copper loss and because rare-earth permanent magnets are inserted in rotor structures such as NdFeB alloys [9]. Because of their merits over other drive motors, such as DCs, inductions [21], and switched reluctance machines [20], PMSMs are commonly used in applications characterized by high performance (e.g., robotics [22], aerospace generators and actuators [23], EVs and HEVs [24]), especially for low and medium power, ranging up to 30 kW [25]. PMSMs may be surface mounted (SM-PMSM) or interior (IPMSM) according to

Table 1 Comparison of electric machines for drive applications

Machine type	Advantages	Disadvantages
DC machine	Easy control Desirable torque speed characteristics	Frequent need of maintenance Low efficiency and reliability Low speed range
Induction machine	High speed range High reliability Low cost	Low power density and large size Low efficiency Thermal problem during high-speed operation
PMSM	High speed range High reliability Low cost	High stator core loss during high-speed operation High cost Uncontrollable PM flux Possible demagnetization effects
SRM	Desirable torque-speed characteristics High reliability Low cost	High torque ripple and noise Low power density Low efficiency

the rotor design configuration. IPMSMs have more advantages over SM-PMSMs, such as higher torque provided by the saliency design of the machine, faster dynamic response, rugged and robust construction, and higher efficiency over a wider speed range for the field weakening technique [9]. Because of their merits, IPMSMs are more suitable for electric drive systems, which are the tested machines in this paper. Reference [26] reviewed the permanent magnet synchronous motor advancement for variable-speed drives and servo applications to the great parallel developments in power electronics and high-energy permanent magnet materials. A PMSM with a variable frequency inverter and a shaft position sensor does not need a squirrel cage winding for starting since it is always synchronized. Then, equivalent circuit models for these motors with high-energy rare-earth magnets are presented. The computed parameters of the fully inset magnet motor (IPMSM) and noninset magnet motor (SM-PMSM) were compared with the measured parameters. Accurate results based on experiments on two laboratory motors were obtained. Reference [27] states that permanent magnet machines are more attractive than induction machines for servo drive applications. PMSMs have sinusoidal back-emfs (sinusoidal flux distribution) and mandate sinusoidal stator currents for the production of constant torque. The PMSM dynamic model and electrical equivalent circuit are presented. BDCMs have trapezoidal back-emfs (trapezoidal flux distributions) and require rectangular stator currents to produce constant torque. The BDCM equations are given in a three-phase system, and there is no advantage in converting the three-phase equations to the d-q axes because the nonsinusoidal variability of the mutual inductances among the stator and the rotor, as in the case of the back-EMF, is not sinusoidal. The experimental results for the two types of PM machines are given and discussed. P. Pillay and P. Freere [28] presented a survey study on permanent magnet (PM) machine design, control, characteristics, and drive systems. PM machines are classified as permanent magnet synchronous machines (PMSMs), which may be self-starting, or as brushless DC machines (BDCMs), which are dependent on variable frequency inverters because currents must be trapezoidal waveforms. The PM machine classifications are shown in the following chart in Fig. 2.

In 1998, reference [29] proposed two improved flux weakening schemes because a wide speed range beyond the base speed is an important requirement of PMSM for servo and electric vehicle applications. The SM-PMSM is considered an

Fig. 2 Inverter PM machine categories

unsuitable machine for this purpose due to the large magnetizing current required for flux weakening, which causes a lack of efficiency at high speeds. The proposed schemes offer an operation characterized by stability and reliability up to 3.5 times the base speed for the SM-PMSM without being dependent on the parameters of the machine. In 2001, reference [30] proposed an accurate method to calculate the IPMSM parameters without torque measurement, including the core losses based on the current, speed, and rotor position. IEC standard recommendations for induction and standard synchronous motor parameter calculations are not useful for PMSMs. Based on the motor parameters, other effects on motor operation, such as the effect motor parameters and frequency dependency, can be studied. In 2010, reference [31] proposed a flux-weakening control technique for the SM-PMSM considering the stator resistance voltage drop. An assessment and discussion of the motor's equations for optimal torque-speed bounds in the constant-power region were presented. A simplified feedforward flux-weakening approach with the characteristics of a fast transient response and the option of introducing optimal techniques that are useful for microcontroller SM-PMSMs was presented. Position tracking application experimental results are given for a speed range of (2–3) as the rated speed.

2.4 PMSM control techniques

The proper control technique is the second factor in accessing a high-performance traction system during different modes of operation. Many different control methodologies, such as field-oriented control (FOC) [25], direct torque control (DTC) [32], and other nonlinear techniques, including sliding mode control (SMC) [33], the fuzzy control technique [34] and neural network techniques [35], have been proposed and implemented for PMSM drives to achieve high-performance operation.

2.4.1 Vector control of PMSM

FOC is a simple technique that treats a three-phase PMSM drive system as a separately excited DC motor [25] by separating the two overlapped orthogonal d-q axis currents. In 1987, P. Pillay and R. Krishnan [25] reviewed the drawbacks of DC servo motor drive compared to AC motor drives. The PMSM nonlinear model is presented with several assumptions for simplicity, such as neglecting the saturation, not having an eddy and having a hysteresis current, and having a sinusoidal and no damper winding. The field-oriented control (FOC) or the vector control that makes the PMSM like a separately excited DC machine by allowing the independent control of the d- and q-axis currents is discussed and applied. The FOC consists of two cascaded control loops, the outer loop, and another inner control loop. The outer loop is for speed control, where the error between the command speed and the measured speed is inserted into a PI controller that produces the command torque used to obtain the reference q-axis current required for the inner q-axis loop. The inner control loops are for current regulation according to the reference values, such as when the constant air gap flux mode d-axis current is zero and the q-axis current is calculated from the command torque. The inner loops are generated by the hysteresis current controller or via the PWM technique. The experimental results are given. In 1990, Rached Dhaouadi and Ned Mohan [36] reviewed the SM-PMSM drive system of machine models, current controllers, and speed controllers with three various types of current controllers. The prime objective for a PMSM is achieving torque control above the desired speed range with an adequate ripple in the output torque waveform. They investigated generating a smooth torque at low speeds and the system operating limits in the high-speed ranges (constant power region) or the flux weakening region. Three different current controllers, the hysteresis controller, the ramp comparison controller, and the space vector controller, were reviewed. The simulation results are given and compared. In 1994, reference [36] mentioned the common use of permanent magnet synchronous motors (PMSMs) for applications in which motor efficiency is a serious feature (e.g., compressor drives and electric vehicle applications over DC and induction machines). Then, they discussed one of several control technique applications for enhancing the performance of PMSMs by controlling the d-axis current according to the operating speed and conditions of the load. The control technique targets the maximization of PMSM motor efficiency by minimizing machine losses. The machine losses are mechanical losses and electrical losses; the latter include copper losses and iron losses, which are the summation of the eddy and hysteresis losses. The mechanical losses are dependent on the machine speed and are not controllable, while the electrical losses are controllable and may be minimized according to the operating speed and the conditions of the load by the armature current vector optimal control technique. The minimization of copper losses is possible through the maximum torque per unit ampere control method, in which the maximum allowable torque can be achieved with the minimum allowable current by controlling the armature current vector. The iron losses are proportional to the flux density squared so that they can be minimized by reducing the flux density, known as the flux-weakening control technique. The PMSM d-q model including the core

losses is presented. The maximizing efficiency control technique is discussed, simulated, and applied experimentally with a digital signal processor (DSP) for a prototype interior permanent magnet synchronous motor. This technique is applicable for surface-mounted permanent magnet synchronous motors (SM-PMSMs) but also has reduced performance because of the saliency and reluctance torque in these machines. In 1997, Ajith H. Wijenayake and Peter B. Schmidt [37] described a d-q axis equivalent circuit model for IPMSM and SM-PMSM by taking iron losses and magnetic parameter variations into account to obtain a more accurate model, making predictions for the machine parameters more accurate, especially for sensorless control schemes in which almost all of them depend on the measurement of the motor currents, voltages, and motor parameter variations that affect the accuracy of the machine model and thus motor performance. To obtain a more accurate model based on no-load parameters and saturation levels, an online parameter identification technique is discussed in detail. The distribution of the airgap flux density at heavy load levels in interior permanent magnet machines is distinctly nonsinusoidal, leading to less performance than that of surface mounted machines. Experimental test schemes for measuring these equivalent circuit parameters and calculating saturation constants that affect parameter variability were presented. In 2021, reference [38] proposed a novel scheme for space vector pulse width modulation (SVPWM) to reduce the problems of systems fed by multilevel inverters, such as computational complexity, high common mode-voltage, and high switching loss. The proposed scheme applied for PMSM FOC is practical and independent of the inverter level. The simulation and experimental results are verified. A lookup table (LUT) [39] based on sliding mode vector control for the EV drive system is suggested and provides better functionality than the conventional PI-controller-based system. The SMC-based fuzzy control technique [40] with load torque estimation for PMSM speed control is designed to guarantee robust speed control under load variations but not SMC.

2.4.2 Direct torque control of PMSM

According to the torque and stator linkage flux control, reference [41] proposed a wide speed range for IPMSM DTC control with a speed sensor by applying an LUT for MTPA operation, while the field weakening operation was based on a reverse relation between the speed and the voltage vector. The DTC enhances dynamic characteristics but provides additional current and torque ripple distortion [42], which is a major problem in sensorless estimation.

2.4.3 Model predictive control of PMSM

Model predictive control (MPC) is a commonly applied technique for PMSM drives and power electronic converters. A detailed MPC for power electronic converter control is discussed and applied for different applications of SM-PMSMs [43] and matrix converters [44]. In advance, reference [10] proposed an integral finite set-MPC for the SM-PMSM with constraints based on the state-space model linearization that was modified to include the field weakening operation in [45]. In [46], a fixed switching frequency MPC for MPC based on a three-level T-type inverter was proposed and validated experimentally. MPC is an alternative control technique to conventional cascaded PI controllers and depends on minimizing a cost function to determine the optimal next control action applied to the motor. The MPC technique was not popular for drive applications because of the requirements for fast sampling and many calculations until the current advancements in power electronics and microcontrollers. The MPC technique can be constrained and extended to handle periodic disturbances. In 2009, reference [47] presented an explanation of the finite control set-model predictive control (FCS-MPC) applied to power electronic converters for different applications of PMSMs, IMs and matrix converters. The FCS-MPC technique is simple and applicable for different converters without the need for modulation techniques or internal cascaded control loops. The limitation of high calculation power requirements has been overcome due to the advancements in DSPs and microcontrollers, so these devices can be used for complex applications such as matrix converters. Simulation and experimental results were given and discussed. In 2014, reference [48] proposed and implemented MPC with a fixed switching frequency (MPC-CSF) for SM-PMSMs using a field programmable gate array (FPGA). The authors subsequently stated that MPC-CSF achieves the same dynamic performance as FCS-MPC. The experimental results show that MPC-CSF has good performance for SM-PMSM drives. In 2016, reference [49] presented a model predictive control (MPC) technique allowing field weakening of the SM-PMSM by linearization of the machine model. An experiment is considered for verifying the proposed algorithm through simulation. In 2017, reference [50] reviewed PMSM drive control techniques; among these techniques, model predictive current control (MPCC) is considered a high-performance control technique because of its quick response and simple principle. The MPC technique selects the best voltage vector that minimizes the current error between the command and the measured values based on a quadratic cost function. The authors proposed two methods to improve the PMCC for PMSM drives by estimating the extended EMF from the

past step and then applying this EMF in the next step of stator current prediction. Simulation and experimental results verify the effectiveness of the two proposed methods in achieving good steady-state performance with rapid dynamic response. In 2018, reference [51] proposed a compensated MPC technique with simplified repetitive control (MPC-SRC) for improving the robustness of MPC against current distortion, parameter mismatch, and deadtime effects for SM-PMSM drives. In the proposed technique, open-loop SRC-two controllers with first and second resonant units are used, and the design of the output filter is ensured to offer the system stability in addition to correcting the quantizing error frequency. The results of the simulation and experiment serve as proof of the substantial improvement in performance of the proposed robust technique and provide a 30% reduction in the THD in the stator phase current compared to that of the conventional MPC technique. In 2020, reference [52] proposed a modulated model predictive control technique based on a two-step (current and voltage) cost function for improving the steady-state performance and reducing the switching losses for SM-PMSM drive systems. The proposed method is verified through simulations and experiments. In 2021, reference [53] proposed an improved model predictive current control (MPCC) technique to improve dynamic and steady-state performance based on the estimation of the back EMF from the previous stator voltage and current vectors to predict the next current vector. The proposed MPCC technique chooses the optimal voltage vector according to the current track circle rather than the cost function for simplicity and quick validation of the results via simulation and experiments. In 2021, reference [54] adopted the FCS-MPC for the realization of the q-axis current three-level NPC inverter fed PMSM with suppression of the LC filter resonance. In 2022, reference [55] proposed a computationally efficient FCS-MPC for multiphase PMSM drive systems to eliminate low-order harmonic current in open-loop mode, which results in simplification of the prediction mode and the cost function. In 2022, reference [56] proposed an enhanced FCS model predictive direct speed control (MPDSC) to improve the robustness of the (internal & external) parameter estimation. The proposed technique abandons the conventional PMSM cascaded control scheme with a slightly faster speed response, and the load disturbance is assessed by a model reference adaptive system (MRAS). In general, Table 2 summarizes a comparison between different control techniques for drive applications [42, 57, 58].

2.5 Senseless control techniques

Providing accurate rotor speed and position is important for attaining a high-performance drive system. Various sensorless speed/position estimation schemes have been proposed for eliminating speed sensors without affecting drive system performance to provide exact rotor speed and position. Speed estimation and control are essential for low-end applications (washing machines and air conditioners) [11] and for applications where safety requirements are functional, such as EV systems [12]. Estimating the speed has many advantages, such as eliminating the speed sensor and its cable, less maintenance, greater robustness, and reduced cost [13, 14]. Over the last few decades, many estimation techniques, such as adaptive nonlinear flux observers [59], Kalman filters [60], model reference adaptive systems (MRASs) [61, 62], sliding mode observers [63], and back-emf estimation [64], have been proposed and implemented. In [65], an extended nonlinear sensorless speed observer for SM-PMSM drives was proposed with load torque estimation based on feedback linearization of the nonlinear model. An indirect FOC technique for an SM-PMSM drive was proposed [66] to estimate the motor speed, permanent magnet rotor flux, and stator resistance based on two MRAS observers. In [67], according

Table 2 Comparison of PMSMs' control techniques for drive applications

Control Technique	Merits	Demerits
FOC	Easy to design Easy to implement in practical applications	Long settling time Large overshoot High speed fluctuation
SMC	Easy to design and implement Low steady state error	Insufficient settling time and convergence rate
DTC	Don't require the use of an inverter with PWM High quality dynamic characteristics No need for reference frame transformers	High ripple levels of current, linkage flux and torque waveforms Less sensitive to motor parameters change and disturbances
MPC	Simple structure Ensuring precise and rapid regulation of motor speed	Dependent on motor model parameters Weighting factors design

to the stability theorem, two sensorless MRAS techniques based on Lyapunov [68] and Popov [69] stability theorems for PMSM drive parameter estimation of stator resistance, inductances, and rotor flux were implemented. The Popov stability estimators depend on the proportional and integral gains and do not need the expertise required in the criterion based on the Lyapunov stability; therefore, these estimators are superior for response speed and convergence speed [67]. In [69], an online algorithm based on an extended Kalman filter was presented to detect and isolate faults in an IPMSM drive system sensor to overcome the system randomly. Back-emf sensorless estimation is not appropriate for low-speed operation because the motor speed is proportional to the emf, which is very low at low speeds, resulting in inaccurate estimation and zero emf during standing, which prevents the estimation process from being applicable. In 2013, reference [70] presented a status review for brushless PM machine sensorless control techniques, including machine design influences on the performances of these sensorless techniques. Permanent magnet machine control techniques require the rotor position required for coordination transformations, which is provided by a position sensor or a speed encoder. However, these techniques suffer from common disadvantages, such as sensitivity to mechanical stress, reduced overall drive system reliability, and high cost.

Current drive research aims to achieve high adjustable speed drive performance with reduced cost by avoiding position sensors, cabling devices, and connectors. The ability of the permanent magnet machine to perform sensorless operation is analyzed, and the optimal design of the machine is determined to provide the best results and performances. Sensorless techniques are classified into two categories according to the applied excitation signal frequency, as shown in Fig. 3. Techniques based on the fundamental frequency commonly used for non-salient machines. However, in some applications, such as automotive and traction majors, position sensors cannot be fully removed, and sensorless techniques are still applied to determine the rotor position and speed when sensor failure occurs.

In 2013, reference [71] proposed a quadrature PLL for rotor speed/position estimation based on the SMO under the MTPA mode of operation for IPMSM. In 2016, reference [72] proposed an STA-SMO for speed estimation with online stator resistance estimation for an SM-PMSM with a zero d-axis current. In 2017, reference [73] studied and implemented sensorless speed estimation for SM-PMSMs based on SMOs under SMC and compared the results of SMOs for MRAS observers. In 2018, reference [74] presented a comparative study of the nonlinear observers of the extended Kalman filter (EKF), the resilient extended Kalman filter (REKF) and the unscented Kalman filter (UKF) for sensorless direct torque (DT) and flux control techniques for SM-PMSM drives. The simulation and experimental results demonstrated the superiority of the resilient extended and unscented Kalman filters over the traditional extended Kalman filter for sensorless-DTC applications for PMSM drives, where REKFs are preferred when sensors fail to provide accurate measurement values and UKFs are superior in estimating the state variables when all sensors give accurate measurement values. In 2018, reference [75] designed a model reference adaptive control-based-speed controller (MRAC-SC). The proposed technique consists of two control parts: an adaptive compensating control term to compensate for uncertain parameters (inertia, viscous friction, and load torque) and a stabilizing feedback control term for stabilizing the dynamic error. A comparative study between the proposed adaptive technique (MRAC-SC), a nonadaptive technique and conventional PI controllers demonstrated the superiority of the MRAC technique for rapid dynamic response and robustness against parameter uncertainties. In 2018, reference [76] proposed multiparameter (speed/position, stator resistance and stator flux) estimation through the use of the Model Reference Adaptive System (MRAS) for the reduction of the negative impact of parameter variability in position sensorless speed control depending on stator-feedforward-voltage estimation (FFVE), as this model technique depends on motor parameters with a Phase Locked Loop (PLL) for speed estimation. The proposed sensorless scheme is similar to the classical sliding mode observer (SMO) but exhibits greater stability, as shown experimentally at low speeds under parameter detuning. The experimental results show the efficiency of the suggested scheme for the

Fig. 3 Classifications of sensorless control techniques for PM machines

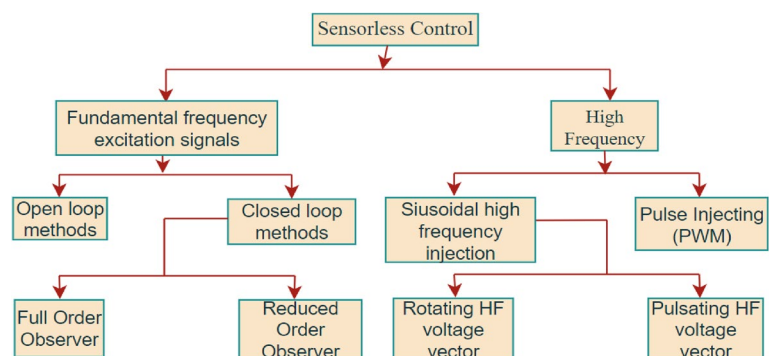
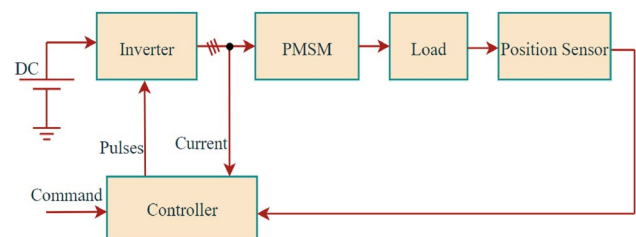


Table 3 Comparison for sensorless observers for speed/position estimation of PMSM's drive systems

Observer	Observer method	Merits	Demerits
Open loop estimation	Non	No tunable parameters	Slow dynamic response Poor robustness Sensitive to disturbances
State observers	MRAS	Multi-parameter estimation Wide operating range	Time consuming gain tuning
	SMO	Fast convergence Robust technique	Chattering problem Complex high-order SMO gain design
	EKF	Anti-measurement noise capability	Computational burden
Signal injection	HF&LF injection	Simple implementation	Voltage and current distortions by high-frequency and dc injection

Fig. 4 Complete Drive System of PMSM

SM-PMSM under variable load and speed conditions. In 2019, reference [77] proposed an adaptive STA-SMO algorithm based on backstepping control with integral action to enhance the speed estimation performance of an IPMSM drive system. In 2019, reference [78] proposed a third-order super twisting extended-state SMO for sensorless rotor speed/position estimation of IPMMs. In 2020, reference [79] reviewed sensorless control techniques for PMSM. A sensorless control technique based on the Model Reference Adaptive System (MRAS) for PMSM is proposed and implemented for the estimation of the rotor speed and position based on the difference between the q-d axis-measured and estimated currents. Experimental results were obtained and discussed, revealing good current control results in the acceleration section and good sensorless speed control performance during low-speed operation. In 2021, reference [80] presented a review on high-speed estimation observers for high-performance sensorless control of PMSM drives. Newly proposed speed observers tend to be applicable and robust against distortions and parameter variations with a low sampling frequency-to-speed ratio, and these factors are discussed. Future trends in the use of sensorless observers for high-speed PMSMs during flux-weakening operations are discussed. In 2022, reference [81] proposed STA-SMO with a complex vector generalized integrator for sensorless speed control for IPMSM considering flying starts based on a virtual resistance that does not affect the speed estimation accuracy. Various sensorless observers' characteristics are summarized in Table 3 [79].

3 Modeling and analysis of the PMSM propulsion drive system

The first step in designing and implementing an IPMSM control system is to construct a mathematical model for the drive system components. The main components of a traction IPMSM drive system are the power electronic inverter and the IPMSM driver motor. This section presents the complete drive system of an IPMSM, as shown in Fig. 4. Then, the mathematical models of the IPMSM and the inverter are presented in the space vector.

3.1 Motor drive system

The drive system is made of an IPMSM, a load fixed on the motor shaft, a DC supply, the power inverter that feeds the IPMSM with the command three-phase voltage, and the controller that depends on a position sensor in the conventional techniques and a sensorless technique in advanced control techniques for determining or estimating the motor speed and the rotor position required for the control system. The following procedures discuss the drive system in detail.

3.1.1 Permanent magnet synchronous machine

AC drive systems are common for mills, motor propulsion, aircraft engines, electric vehicles, subway transportation, home appliances, wind generation systems, services, robotics, etc. PMSMs use permanent magnets in the rotor structure to generate an air gap magnetic field instead of using electromagnets. PMSMs have become more common for AC drive systems in the small power range (1–10 kW) [6, 82], for which up to 30 hp applications are currently available for servo applications [25]. Other high-power ranges exist for wind power generation and ship propulsion.

3.1.2 Types of permanent magnet synchronous motor

The most common permanent magnet motors for ac drives are the radial field PM types used for axial field motors [19]. The radial PM motors are classified as surface mounted (SM-PMSM) or interior (IPMSM) according to the rotor PM set structure.

3.1.2.1 Surface-mounted permanent magnet synchronous motor SM-PMSMs have PMs mounted on the surface of the rotor, as shown in Fig. 5 (a). SM-PMSMs are easy to build, and the particularly skewed poles are easy to use in terms of magnetization to minimize the cogging torque. SM-PMSMs are used for low-speed operation to prevent magnets from flying apart during high-speed operation [3].

These motors have equal d-q axis inductances ($L_{ds} = L_{qs}$) [8] because the permeability of the magnets is almost the same as that of air. For the SM-PMSMs' d-axis (field excitation), the current is usually kept at zero (constant flux operation) for maximum efficiency [29].

3.1.2.2 Interior/buried permanent magnet synchronous motor IPMSMs have buried permanent magnets inside the rotor structure, as shown in Fig. 5.b. These buried magnets result in an inductance variation. This inductance variation is a result of the permanent magnet permeability, which is the same as that of air, so it extends the air gap inside the rotor structure. Reference [3] described many categories of buried PMSMs according to the PM configurations inside the rotor. According to the topology illustrated in Fig. 5.b, a steel permeability larger than that of air results in a q-axis inductance higher than that for the d-axis ($L_{ds} < L_{qs}$) [8]. The buried magnets give more merits to the IPMSM, such as higher torque, known as reluctance torque [25], and high-speed operation. For the IPMSM, a nonzero field excitation d-axis current is applied to obtain an optimized torque per ampere during low-speed operation and to weaken the field during the field weakening operation to obtain a higher speed than the base speed, as shown in the following sections. IPMSMs have merits over SM-PMSMs, which can be summarized as in Table 4.

3.2 Voltage source inverter modeling and control

The efficient control of power electronic converters and machine drives is achieved by on/off control of semiconductor devices. On/off switch control manipulates the command control signals in three-phase form. The duration of on/off semiconductor switches depends on the command voltage signals. There are many different converters' technologies. Between these various technologies, two-level voltage source inverters (2 L-VSIs) and matrix converters (MCs) are most common for variable speed drives [83].

Fig. 5 PMSM structure, **a** SM-PMSM type, and **b** IPMSM type

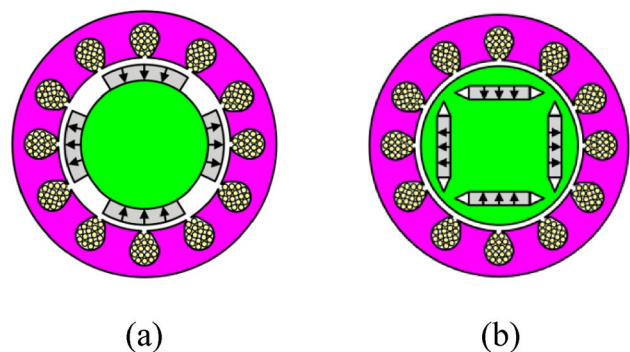


Table 4 Comparing SM-PMSMs to IPMSMs

SM-PMSMs	IPMSMs
Simple construction	More complex
Low flux leakage	Higher leakage
Medium speed range	Wide speed range
Poor flux weakening capability	High flux weakening capability
	Commonly used

3.2.1 Power matrix converter

Last decades, there is a great research interest on MCs that are single-stage AC-AC power converters capable of connecting the three-phase supply to the load directly without energy storage system. It consists of an array of bidirectional controlled semiconductor switches as shown in Fig. 6. MCs have some desirable features such as [84]:-

- Simple and compact power flow
- Bidirectional power flow
- Deliver voltage/current to the load with high-quality and without restrictions on the frequency

3.2.2 Two-level power inverter modeling

Among the many types of converter technologies, the 2 L-VSI, demonstrated in Fig. 7, is commonly used for variable AC-drive control because it is simple and easy technology. The 2 L-VSC consists of three legs, each of which includes two switches that must not operate simultaneously to prevent short-circuited source circuits. The input signal for each switch to turn it on/off is the firing signal or the gate signal. The most common technique for generating gate firing

Fig. 6 Matrix converter

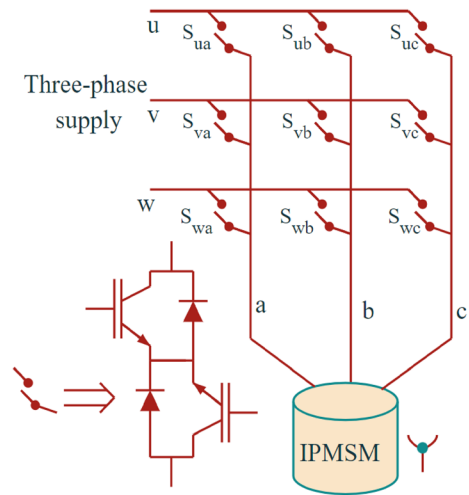
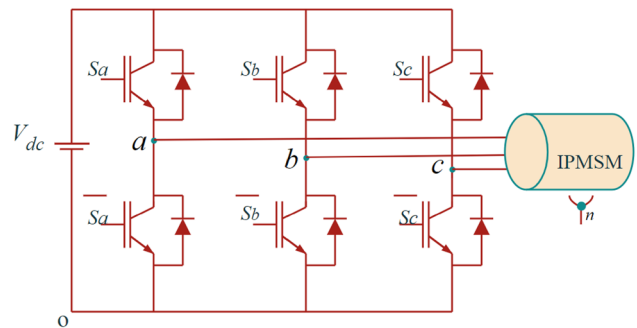


Fig. 7 Two-level voltage source inverter representation



signals for inverter switches is the pulse width modulation technique (PWM). The PWM technique, which depends on the PI and model predictive controllers, is implemented in this paper. The three-phase command input voltage signals of the VSI are obtained from the controller either in d-q or α - β coordinates. The PI-controller PWM technique of the VSI involves chopping the DC-bus signal via the PWM to generate three-phase voltages that are very similar to the controllers' three-phase control signals. The predictive control methodology used to generate the firing signals of the switches involves optimizing the error cost function among the command control signals and the ones attained by the inverter switches, as will be discussed later in the following sections.

The insulated gate bipolar junction transistor (IGBT) is one of the most commonly used power electronics converter switches. IGBTs are integrated with parallel freewheeling diodes to conduct the current in two directions. Three-phase IGBTs 2 L-VSI, shown in Fig. 6, consist of three legs, each of which has two IGBT switches parallel to the diodes. The middle point of each leg is connected to a load terminal of an AC motor. The inverter DC supply (V_{dc}) is divided into two sources, each of which shares half of the total voltage ($V_{dc}/2$). The three-phase voltages are represented with respect to the ground point (o):

$$V_{jo} = V_{dc} S_{j-g} \quad (1)$$

where $S_{j-g} \in \{0, 1\}$ and $j = a, b, c$

Figure 8 shows a simplified equivalent circuit of the VSI with the motor circuit.

The calculations of the VSI output voltages relevant to the neutral point of (°) are provided in (2).

$$V_{jn} = V_{jo} - V_{no} \quad j = a, b, c \quad (2)$$

Based on (2) and considering that the three-phase motor is a balanced load,

$$V_{an} + V_{bn} + V_{cn} = 0 \quad (3)$$

$$V_{no} = \frac{1}{3}(V_{ao} + V_{bo} + V_{co}) \quad (4)$$

$$V_{an} = \frac{2}{3}V_{ao} - \frac{1}{3}(V_{bo} + V_{co}) \quad (5)$$

$$V_{bn} = \frac{2}{3}V_{bo} - \frac{1}{3}(V_{ao} + V_{co}) \quad (6)$$

$$V_{cn} = \frac{2}{3}V_{co} - \frac{1}{3}(V_{ao} + V_{bo}) \quad (7)$$

There are eight different voltage vectors of the VSI according to the eight statuses of the switches. Table 5 shows the voltage vectors of the inverter switches and the output voltage vectors.

Fig. 8 Simplified VSI equivalent circuit

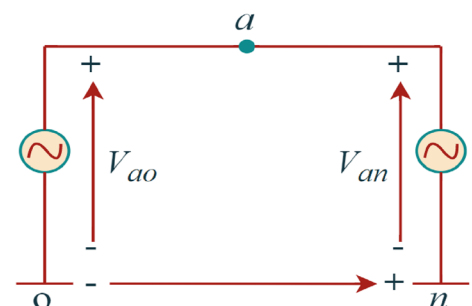


Table 5 VSI switching statuses and voltage vectors

S_a	S_b	S_c	V_{ao}	V_{bo}	V_{co}	V_{an}	V_{bn}	V_{cn}
0	0	0	0	0	0	0	0	0
0	0	1	0	0	V_{dc}	$-\frac{V_{dc}}{3}$	$-\frac{V_{dc}}{3}$	$\frac{2V_{dc}}{3}$
0	1	0	0	V_{dc}	0	$-\frac{V_{dc}}{3}$	$\frac{2V_{dc}}{3}$	$-\frac{V_{dc}}{3}$
0	1	1	0	V_{dc}	V_{dc}	$-\frac{2V_{dc}}{3}$	$\frac{V_{dc}}{3}$	$\frac{V_{dc}}{3}$
1	0	0	V_{dc}	0	0	$\frac{2V_{dc}}{3}$	$-\frac{V_{dc}}{3}$	$-\frac{V_{dc}}{3}$
1	0	1	V_{dc}	0	V_{dc}	$\frac{V_{dc}}{3}$	$-\frac{2V_{dc}}{3}$	$\frac{V_{dc}}{3}$
1	1	0	V_{dc}	V_{dc}	0	$\frac{V_{dc}}{3}$	$\frac{V_{dc}}{3}$	$-\frac{2V_{dc}}{3}$
1	1	1	V_{dc}	V_{dc}	V_{dc}	0	0	0

3.2.3 Power inverter control (firing pulse generation) scenario

The gate signals of the 2 L-VSI can be generated by modulation or pulse width modulation. The commonly used PWM methods include sinusoidal pulse width modulation (S-PWM) and space vector pulse width modulation (SV-PWM).

The maximum amplitude limit of the command three-phase voltages that enter the VSI during PWM is $\frac{2V_{dc}}{\pi}$. To ensure that the operation is within the linear region (the modulation is lower than 1 to not saturate), the maximum limit must not exceed $\frac{V_{dc}}{2}$ to ensure the linear modulation region. For SV-PWM, the linear operating region limit is limited by $\frac{1}{\sqrt{3}}V_{dc}$, where the submodulation range is 90.7%, while the traditional value is 78.5%. The SV-PWM achieves the maximum limit of inverter output voltage utilization, which is 15% greater than that of the PWM [85]. The SV-PWM is the proposed technique for ensuring the maximum utilization of the inverter limits, which makes it more suitable for traction applications than other PWM techniques [86]. To ensure that the modulation is in the linear range, α β reference voltages must satisfy the limits in (8) and the d-q frame limit in (9).

$$\sqrt{(V_{\alpha s})^2 + (V_{\beta s})^2} \leq \frac{V_{dc}}{\sqrt{3}} \quad (8)$$

$$\sqrt{(V_{ds})^2 + (V_{qs})^2} \leq \frac{V_{dc}}{\sqrt{3}} \quad (9)$$

The SV-PWM generates the inverter output voltage based on the principle of SV representation. Figure 9 shows a space vector diagram of the possible eight vectors ($V_0, V_1, V_2, V_3, V_4, V_5, V_6, V_7$) of the inverter output voltages. Two of the vectors are zero vectors (V_0, V_7), and the others are six active vectors. The space vector command voltage may be applied during

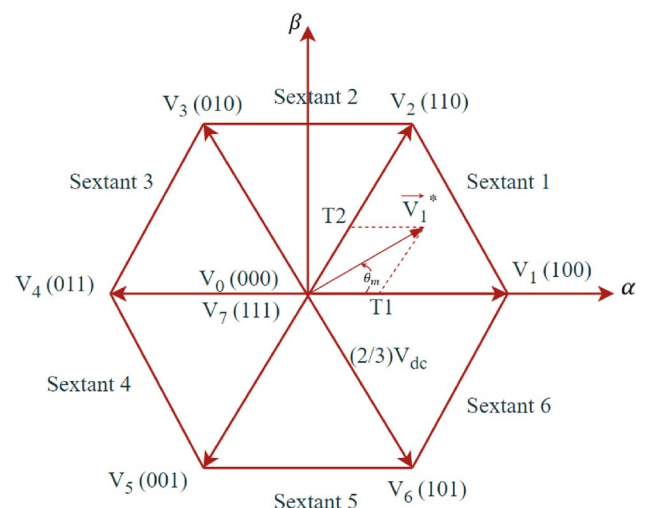
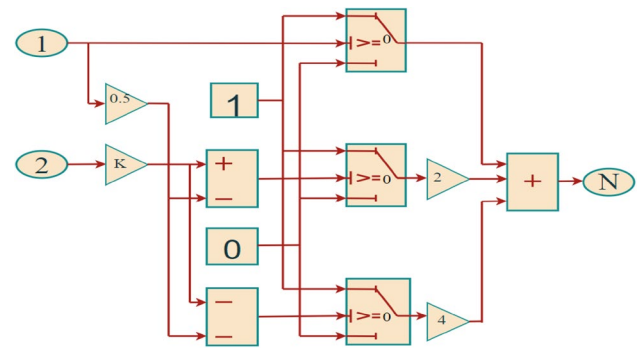
Fig. 9 SV-PWM representation

Fig. 10 Sector identification**Table 6** Shows the sextants vs. number N

Sextant	1	2	3	4	5	6
N	3	1	5	4	6	2

six different sectors or sextants. Vector $\vec{V}^*$ is the command vector to be generated by the SV-PWM principle, where each command vector is a summation of two adjacent active vectors and a zero vector for a specified time T.

3.2.4 SV-PWM implementation on MATLAB-simulink

The main idea is to select the two adjacent active vectors of the command vector plus the zero vector. The implementation of the SV-PWM algorithm consists of the following steps [81]:

4 Identify the working sextant.

Using the command voltage vector in the $\alpha\beta$ reference frame:

$$\text{if } \begin{cases} V_{\beta} > 0 \rightarrow A = 1, & \text{else } A = 0 \\ \sqrt{3}V_{\alpha} - V_{\beta} > 0 \rightarrow B = 1, & \text{else } B = 0 \\ -\sqrt{3}V_{\alpha} - V_{\beta} > 0 \rightarrow C = 1, & \text{else } C = 0 \end{cases} \quad (10)$$

where V_{β} , $\sqrt{3}V_{\alpha} - V_{\beta}$, $-\sqrt{3}V_{\alpha} - V_{\beta}$ are auxiliary space vector reference voltages. Then, the sector N is calculated as:

$$N = A + 2B + 4C \quad (11)$$

Figure 10 shows the Simulink model of sextant identification according to (11) (Table 6).

5 Calculation of the fundamental vector operation time

Depending on the PWM switching period T and fundamental vector operating times (X, Y, Z), the operating time of the two adjacent active vectors (T1, T2) is calculated as follows:

$$\begin{cases} X = \frac{2V_{\alpha}}{\sqrt{2}V_{dc}} T \\ Y = \frac{(\sqrt{3}V_{\alpha} + V_{\beta})}{\sqrt{2}V_{dc}} T \\ Z = \frac{(\sqrt{3}V_{\alpha} - V_{\beta})}{\sqrt{2}V_{dc}} T \end{cases} \quad (12)$$

These operating times are presented in MATLAB-Simulink in Fig. 11 (Table 7).

Fig. 11 Fundamental vector operating times in Simulink

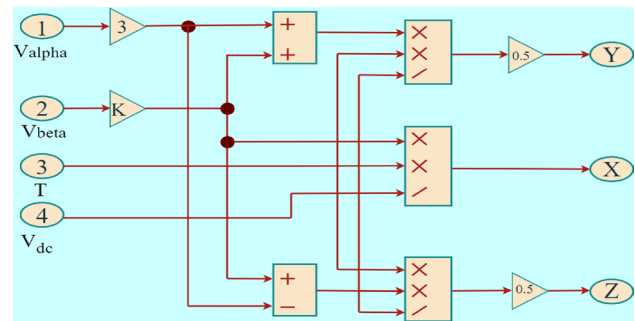
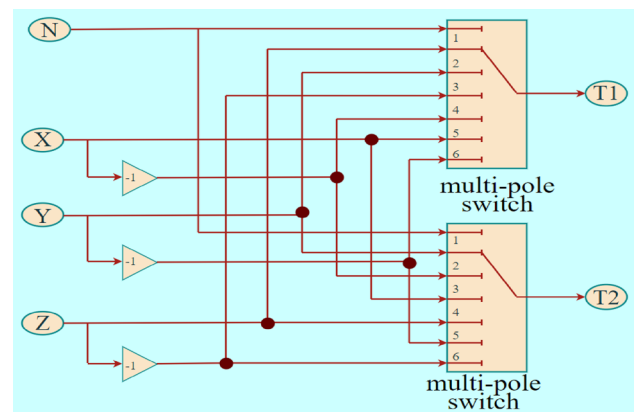


Table 7 Presents the fundamental vector operating times

Sextant	1	2	3	4	5	6
T1	Z	Y	-Z	-X	X	-Y
T2	Y	-X	X	Z	-Y	-Z

Fig. 12 Active vector time calculation



The active vectors are calculated depending on the fundamental vectors are presented in Fig. 12.

Time T1 and T2 are parts of T. Their summation must be smaller than the PWM period time T. For the saturation case, if the summation of T1 and T2 is larger than the period T, then the time parts are updated by taking them as a ratio of the period time as follows:

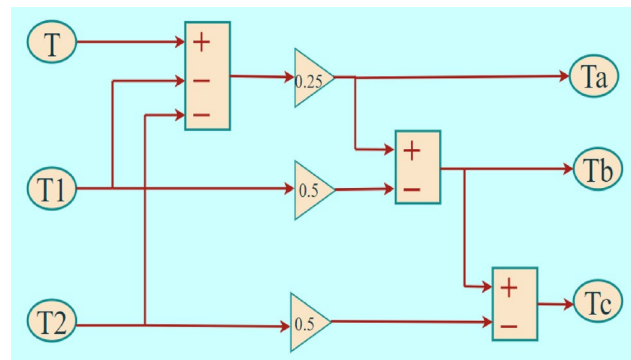
$$\text{if } T1 + T2 > T \begin{cases} T1 = \frac{T1}{T1+T2} T \\ T2 = \frac{T2}{T1+T2} T \end{cases} \quad (13)$$

6 SV-PWM Pluses Generation and Complete Simulink Model Representation

The switching operating times related to the number of sextants (N) are shown in Table 5, and the corresponding simulation model is presented in Fig. 13 (Table 8).

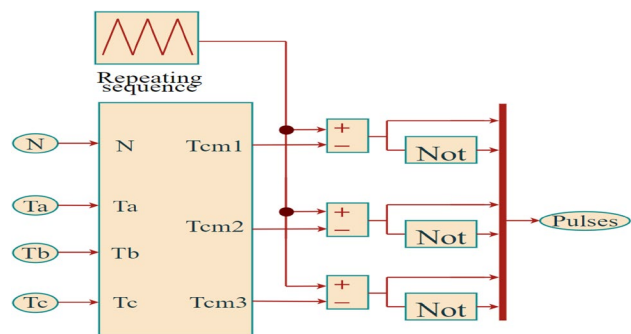
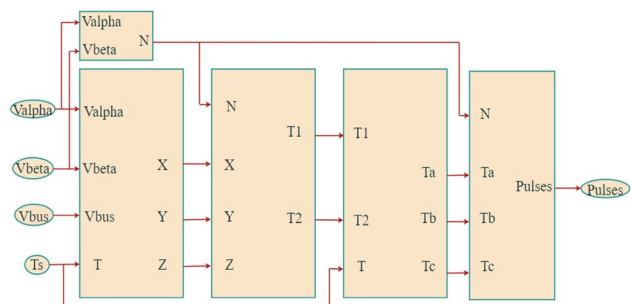
$$\begin{cases} Ta = 0.25(T - T1 - T2) \\ Tb = Ta + 0.5T1 \\ Tc = Ta + 0.5T2 \end{cases} \quad (14)$$

By comparing the calculated three-phase operating times Tcm1, Tcm2, and Tcm3 with the triangle carrier signal, three pulses are generated for three upper switches of the inverter and inverted by no function to obtain the other three pulses, as shown in Fig. 14.

Fig. 13 SV-PWM modulation periods vs. sectors**Table 8** Operating time over different sextants

Sextant	1	2	3	4	5	6
Tcm1	Tb	Ta	Ta	Tc	Tc	Tb
Tcm2	Ta	Tc	Tb	Tb	Ta	Tc
Tcm3	Tc	Tb	Tc	Ta	Tb	Ta

Where: T_a , T_b , T_c are the three phase SV-PWM modulation periods and Tcm1, Tcm2, and Tcm3 are the three phases of operation

Fig. 14 Three-phase operation times for Tcm1, Tcm2, and Tcm3 and for SV-PWM generation**Fig. 15** The complete simulation model of SV-PWM

The complete SV-PWM Simulink model is presented in Fig. 15.

6.1 Permanent magnet synchronous motor model

As discussed in the preceding sections, IPMSMs have more practical use than SM-PMSMs due to their superior advantages. Burying the permanent magnets inside the rotor provides both a mechanical rugged system and an increase in the capability of higher-speed operation. IPMSMs are not only permanent magnet machines but also reluctance

machines that produce two torque terms, as discussed in this section. The main requirements of the IPMSM for EV drive systems are as follows:

1. Improved utilization of VSI-DC supply via the SV-PWM by a 15% increase;
2. Increased motor overall efficiency;
3. High-speed operation capability;
4. Improved constant-power-speed range capability of the IPMSM with the SV-PWM.

To obtain the IPMSM model in the d-q axis rotating reference frame, there are several assumptions [25]:

1. The three-phase PMSM machine is symmetrical;
2. No damper winding;
3. The distributed stator and rotor winding are replaced by a concentrated coil;
4. Hysteresis losses and eddy currents are ignored;
5. Anisotropy effects and magnetic saturation are neglected;
6. The coil resistance and reactance are constant;
7. No thermal effect on permanent magnets;
8. No field current dynamics;
9. Sinusoidal emf.

The simplified and idealized model of an interior permanent magnet synchronous motor is described as three windings in the stator and PMs in the rotor, as illustrated in Fig. 16.

The electrical model of the IPMSM in the d-q rotating reference frame [25] is shown in the following equations:

$$u_{ds} = R_s i_{ds} + L_{ds} \frac{di_{ds}}{dt} - \omega_m L_{qs} i_{qs} \quad (15)$$

$$u_{qs} = R_s i_{qs} + L_{qs} \frac{di_{qs}}{dt} + \omega_m L_{ds} i_{ds} + \omega_m \psi_f \quad (16)$$

$$J_m \frac{d\Omega_m}{dt} = T_{em} - T_l \quad (17)$$

$$\frac{d\omega_m}{dt} = \frac{Z_p}{J_m} (T_{em} - T_l) \quad (18)$$

Fig. 16 PMSM configuration

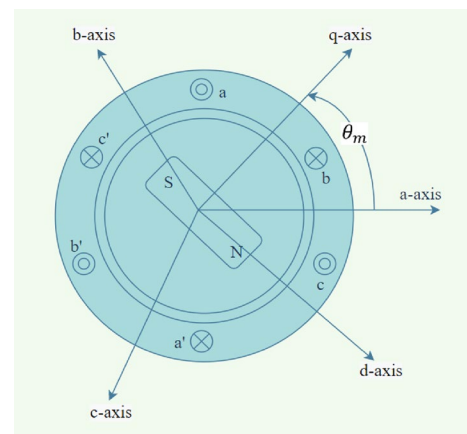
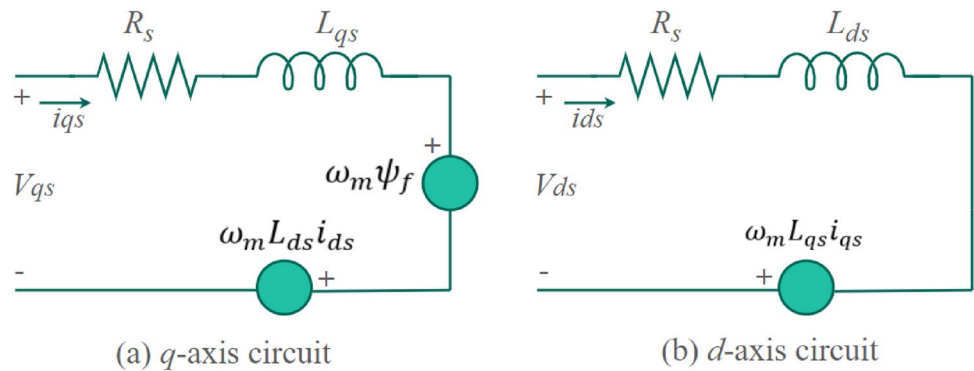
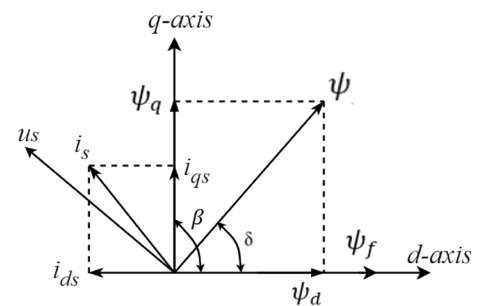


Fig. 17 IPMSM equivalent circuit in the d-q frame**Fig. 18** IPMSM vector diagram

$$T_{em} = \frac{3Z_p}{2} (\psi_f i_{qs} + (L_{ds} - L_{qs}) i_{ds} i_{qs}) \quad (19)$$

where ω_m is the electrical motor speed in electrical rad/s, which is equal to the rotor mechanical speed (Ω_m) multiplied by the pole pairs (Z_p) as in Eq. (20).

$$\omega_m = Z_p \Omega_m \quad (20)$$

T_{em} is the electromagnetic torque and is composed of two terms: the first term is the magnetic torque generated by the permanent magnet flux, and the second term is the reluctance torque generated by the difference in d-q reluctances in the IPMSM according to its design. This reluctance torque is an advantage of the interior machine over ordinary SM-PMSMs. The IPMSM equivalent circuit in the d-q rotating reference frame is shown in Fig. 17.

The IPMSM input power in the three-phase and d-q reference frames is as follows in Eq. (21):

$$P_{in} = [u_{as} \ u_{bs} \ u_{cs}] \begin{bmatrix} i_{as} \\ i_{bs} \\ i_{cs} \end{bmatrix} = \frac{3}{2} (u_{ds} i_{ds} + u_{qs} i_{qs}) \quad (21)$$

The phasor diagram of the IPMSM is illustrated in Figure 18, and the parameters of the IPMSM used in this paper are illustrated in Table 10, appendix A.

7 Field-oriented control of PMSM drives

After building the space vector mathematical models of the VSI and its SV-PWM control system and the interior permanent magnet machine model, this section describes the operation of the IPM that includes the maximum torque per ampere (MTPA) and the field weakening (FW) operation at higher speeds. The proposed control techniques are applied, and the results are discussed.

7.1 Operating modes of an IPMSM

The operating modes of an IPMSM are based on the target of operation. Among the different operating modes of PMSMs (e.g., constant torque angle [62], unity power factor [86] and maximum efficiency [87]), there are two common operating modes: constant torque and field weakening [88].

7.1.1 Constant torque angle control technique

Constant torque angle control is achieved by maintaining the angle between the stator phasor current i_s and the rotor d-axis flux at 90° . This means that i_{ds} is equal to zero, as shown in Fig. 19. Torque is generated merely by the quadrature axis current, which equals the whole stator current. The electromagnetic torque in Nm is calculated by substituting the zero direct axis current in the torque, as shown (19), so we obtain the following equation, as shown. (22):

$$T_{em} = \frac{3}{2} Z_p \psi_f i_s \quad (22)$$

This mode is used for energy savings during steady-state operation, but this approach is suitable for the SM-PMSM. If this mode of operation is applied to IPMSMs, reluctance torque will be lost, which causes a reduction in motor efficiency; therefore, this technique is not suitable for interior machine control.

7.1.2 Optimized/maximum torque per unit ampere control

The motor losses include core iron losses, copper winding losses and stray losses. For a speed below the rated speed, the dominant loss is the copper loss, so the objective is to optimize the torque per ampere to reduce the absorbed current for the same load produced at a lower copper loss, higher efficiency and thus higher machine performance. The maximum torque per unit ampere operation requires the reluctance torque to be activated in the interior machines, which is an advantage of this machine. According to (19), to obtain a positive reluctance torque for the IPMSM with ($L_{qs} > L_{ds}$), i_{ds} must be negative. From Fig. 20, the current equations are as follows:

$$i_s = \sqrt{i_{ds}^2 + i_{qs}^2} \quad (23)$$

$$i_{qs} = i_s \sin(\delta) \quad (24)$$

$$i_{ds} = i_s \cos(\delta) \quad (25)$$

Substituting (24) and (25) into (19) gives the following torque equation:

$$T_{em} = \frac{3Z_p}{2} \left(\psi_f i_s \sin(\delta) + \frac{1}{2} (L_{ds} - L_{qs}) i_s^2 \sin(2\delta) \right) \quad (26)$$

Fig. 19 Phasor diagram for a constant torque angle

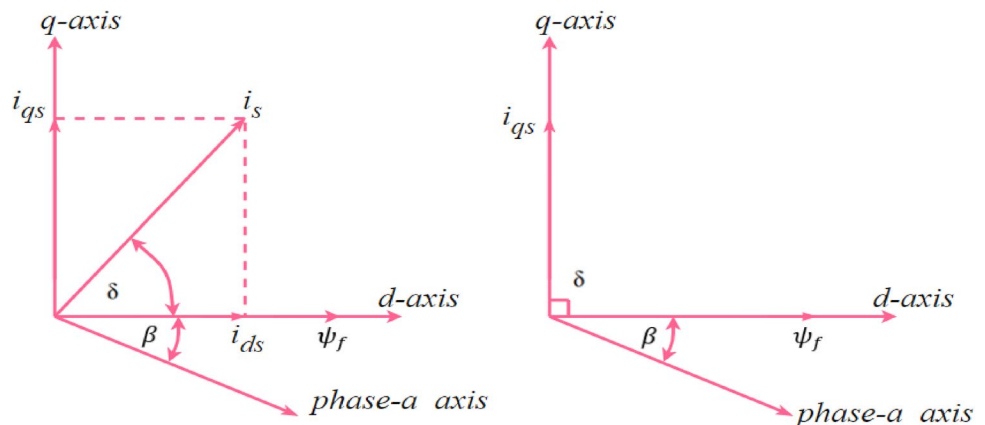
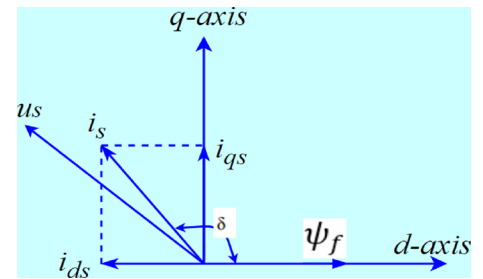


Fig. 20 Stator current phasor diagram



The torque during MTPA mode is constant with respect to the motor speed during all the operating steady-state conditions, as shown in Fig. 21.

Achieving the optimized torque per ampere is possible by differentiating (26) according to δ and setting the result equal to zero. Therefore:

$$\frac{3Z_p}{2} (\psi_f i_s \cos(\delta) + (L_{ds} - L_{qs}) i_s^2 \cos(2\delta)) = 0 \quad (27)$$

$$\psi_f i_s \cos(\delta) + (L_{ds} - L_{qs}) i_s^2 \cos(\delta) - (L_{ds} - L_{qs}) i_s^2 \sin^2(\delta) = 0 \quad (28)$$

Then, a second order current equation can be obtained as follows:

$$\psi_f i_{ds} + (L_{ds} - L_{qs}) i_{ds}^2 - (L_{ds} - L_{qs}) i_{qs}^2 = 0 \quad (29)$$

Solving the second-order current equation produces the command d-axis current corresponding to the optimum torque:

$$i_{ds}^{cmd} = \frac{\psi_f}{2(L_{qs} - L_{ds})} - \sqrt{\left(\frac{\psi_f}{2(L_{qs} - L_{ds})}\right)^2 + i_{qs}^2} \quad (30)$$

The MTPA current trajectory is shown in Fig. 22.

Substituting (19) gives the relation between the command optimum torque and the command quadrature axis current i_{qs}^{cmd} as follows:

$$T_{em}^{cmd} = \frac{3Z_p}{2} i_{qs} \left[\frac{\psi_f}{2} + \sqrt{\left(\frac{\psi_f}{2}\right)^2 + ((L_{qs} - L_{ds}) i_{qs})^2} \right] \quad (31)$$

Using (31) and known values of the machine torque limits, the corresponding q-axis current can be calculated. All these values are placed in the LUT illustrated in Table 9 to directly calculate the command q-axis currents for different command torques. Finally, the command d-q axis currents required for the inner loop are calculated by (30) and the organized LUT. For the IPMSM parameters, see Table 10 in appendix A.

The MTPA range limits are studied during steady-state operation. The steady-state model of the IPMSM is:

Fig. 21 Constant torque operating mode

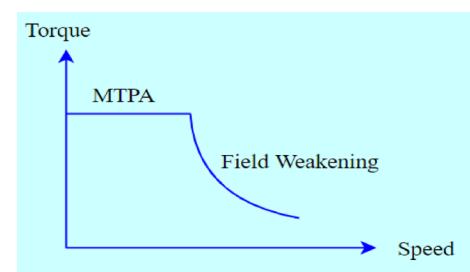
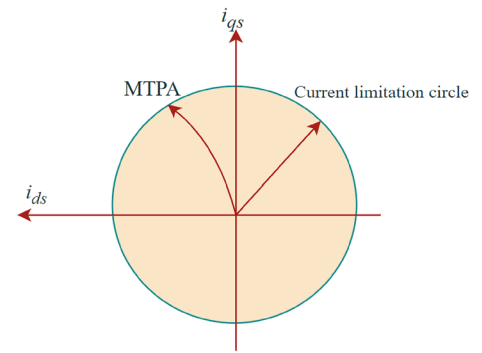


Fig. 22 Current vector trajectory during the MTPA

$$u_{ds} = R_s i_{ds} - \omega_m L_{qs} i_{qs} \quad (32)$$

$$u_{qs} = R_s i_{qs} + \omega_m L_{ds} i_{ds} + \omega_m \psi_f \quad (33)$$

$$\psi_s = \sqrt{\psi_{ds}^2 + \psi_{qs}^2} \quad (34)$$

$$\psi_{ds} = L_{ds} i_{ds} + \psi_f \quad (35)$$

$$\psi_{qs} = L_{qs} i_{qs} \quad (36)$$

$$u_s = \sqrt{u_{ds}^2 + u_{qs}^2} = \omega_m \sqrt{\psi_{ds}^2 + \psi_{qs}^2} \quad (37)$$

$$E_{ds} = \omega_m L_{ds} i_{ds} + \omega_m \psi_f \quad (38)$$

$$E_{qs} = \omega_m L_{qs} i_{qs} \quad (39)$$

Table 9 LUT of the machine command torque and q-axis current

Command torque	0	1	3	4	5	6	7.16	10.65
q-axis current	0	0.85	2.4	3.09	3.7	4.2	4.88	6.486

Table 10 IPMSM parameters [96]

Symbol	Description	Value
Z_p	Pole Paris	2
J_m	Moment of inertia	0.021 kg.m ²
B	Viscous friction	≈ 0
R_s	Stator resistance	1.4852 Ω
L_{ds}	d-axis stator inductance	0.0955 H
L_{qs}	q-axis stator inductance	0.1415 H
ψ_f	Rotor PM flux	0.3847
ω_m	Rated speed	2000 rpm
T_{em}	Rated Electromagnetic torque	7.162 Nm
V_{dc}	DC-Link voltage	540 V
i_s	Phase current amplitude	7.4 A
P	Rated power	1.5 kW
T_{em_max}	Max. Electromagnetic torque	9.936 Nm

$$u_{smax} \geq \sqrt{u_{ds}^2 + u_{qs}^2} \quad (40)$$

$$i_{smax} \geq (i_s \sqrt{i_{ds}^2 + i_{qs}^2}) \quad (41)$$

where E_{ds}, E_{qs} corresponds to the stator d-q-induced electric-motive forces proportional to the motor speed times the stator d-q-axis field, known as cross-coupling terms. Equations (40) and (41) are the constraints of machine operation according to the VSI limits, while $u_{smax} = \frac{V_{dc}}{\sqrt{3}}$ for the SV-PWM is as discussed in Sect. 3, i_s is the stator current of the motor, and i_{smax} is the peak magnitude of the supply current. During steady-state operation limits, neglecting the stator resistance, (37) becomes:

$$\left(\frac{u_{smax}}{\omega_m}\right) \geq \sqrt{(\psi_f + L_{ds}i_{ds})^2 + (L_{qs}i_{qs})^2} \quad (42)$$

Then, the speed formula is:

$$\omega_m = \frac{u_{smax}}{\sqrt{(\psi_f + L_{ds}i_{ds})^2 + (L_{qs}i_{qs})^2}} \quad (43)$$

Then, the base speed for the MTPA from the last equation based on the limits of the VSI is:

$$\omega_m^{base} = \frac{u_{smax}}{\sqrt{(\psi_f + L_{ds}i_{ds_{max}})^2 + (L_{qs}i_{qs_{max}})^2}} \quad (44)$$

7.1.3 Field weakening operation

The motor operation for AC drives requires a higher speed than the rated speed, which requires another technique for controlling the MTPA. Beyond the base speed based on the machine parameters, a higher speed is captured by another technique other than the MTAP known as the field weakening (FW) control technique. This technique is based on reducing the torque of the machine so that it can produce higher speeds for the same power rating. Figure 23 shows the operating modes of the interior machine of the MTPA mode and thus the higher-speed operation. A higher-speed operation is considered to constitute a constant-power operating region.

Achieving higher motor speed without saturating the machine is possible by reducing the flux, which is known as the field weakening operation. The FW technique requires a higher negative d-axis current than does the MTPA and a reduced q-axis current to obtain the maximum torque required during FW operation. For the critical constrained point (with no stator resistance):

$$\left(\frac{u_{smax}}{\omega_m}\right)^2 = (L_{ds}i_{ds} + \psi_f)^2 + (L_{qs})^2(i_{smax}^2 - i_{ds}^2) \quad (45)$$

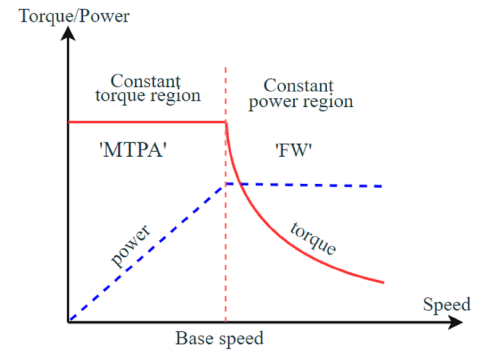
Then,

$$(L_{ds}^2 - L_{qs}^2)i_{ds}^2 + 2L_{ds}\psi_f i_{ds} + (\psi_f^2 + L_{qs}^2 i_{smax}^2) - \left(\frac{u_{smax}}{\omega_m}\right)^2 = 0 \quad (46)$$

For a required speed, (40) is a quadratic equation of the d-axis current that is solved to give the command d-axis current as follows:

$$i_{ds}^{cmd} = \frac{-2L_{ds}\psi_f + \sqrt{(2L_{ds}\psi_f)^2 - 4(L_{ds}^2 - L_{qs}^2)(\psi_f^2 + L_{qs}^2 i_{smax}^2) - \left(\frac{u_{smax}}{\omega_m}\right)^2}}{2(L_{ds}^2 - L_{qs}^2)} \quad (47)$$

Fig. 23 IPMSM operating modes



Then, (23) gives the command q-axis current as follows:

$$i_{qs}^{cmd} = \sqrt{(i_{s_{max}})^2 + (i_{ds}^{cmd})^2} \quad (48)$$

Using the command d-q axis currents, the maximum torque calculation is performed and then compared to the command from the PI speed controller. The smaller value is chosen according to (49) and (50) [89]. Then, the q-axis current is updated from the selected torque via (19).

$$T_{em}^{max} > T_{em}^{cmd} \rightarrow T_{em} = T_{em}^{cmd} \quad (49)$$

$$T_{em}^{max} < T_{em}^{cmd} \rightarrow T_{em} = T_{em}^{max} \quad (50)$$

7.2 Field-oriented control of an IPMSM

Figure 24 shows a schematic diagram of the field-oriented control method. The FOC technique consists of two cascaded control loops with three PI controllers. The outer loop is used for speed regulation, producing the reference currents required for the inner control loops according to the operation mode.

7.2.1 Inner control loop design

By neglecting the cross-coupling compensating terms that are added after designing the PI controller, as shown in the control loop in Fig. 25 and the complete control schematic diagram in Fig. 25, the following voltage equations are obtained:

$$v_{ds} = R_s i_{ds} + L_{ds} \frac{di_{ds}}{dt} \quad (51)$$

$$v_{qs} = R_s i_{qs} + L_{qs} \frac{di_{qs}}{dt} \quad (52)$$

Fig. 24 Inner control loop schematic diagram

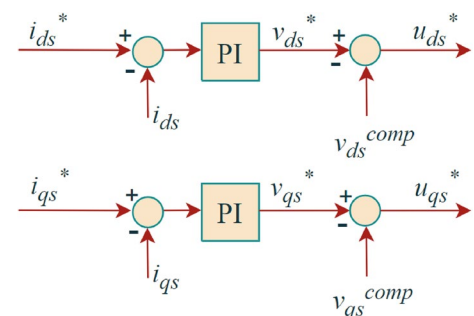


Fig. 25 Schematic diagram of FOC for an IPMSM

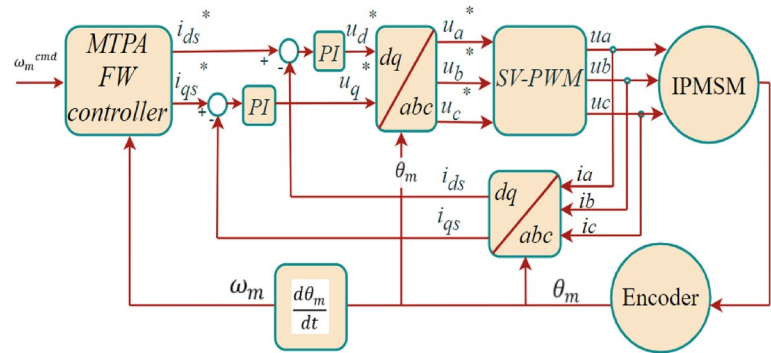
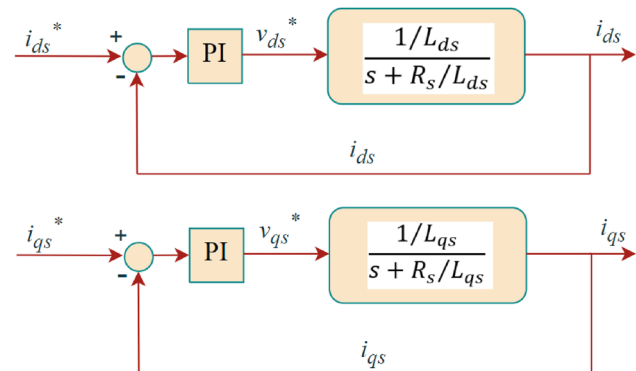


Fig. 26 Simplified d-q axis current loops



Applying the Laplace transform to voltage (51), (52) gives the simplified transfer functions of the d-q axis shown in the simplified control loop in Fig. 26 with a unity VSI transfer function. In the S-domain, the steady-state d-q axis transfer functions of the IPMSM and the PI controller are:

$$\frac{i_{ds}(s)}{v_{ds}(s)} = \frac{1/L_{ds}}{s + R_s/L_{ds}} \quad (53)$$

$$\frac{i_{qs}(s)}{v_{qs}(s)} = \frac{1/L_{qs}}{s + R_s/L_{qs}} \quad (54)$$

Equation (55) is the PI-controller transfer function, where K_p and K_i are the proportional and integral gains of the PI-controller, respectively.

$$PI(s) = K_p + \frac{K_i}{s} \quad (55)$$

Equation (56) gives the closed loop transfer function based on the controller gains. Comparing (56) to the standard second-order transfer function in (57) gives the controller gains in (58).

$$CLTF = \frac{bK_p s + bK_i}{s^2 + (a + bK_p)s + bK_i} \quad (56)$$

$$TF = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (57)$$

$$K_p = \frac{2\zeta\omega_n - a_{d,q}}{b_{d,q}}, K_i = \frac{\omega_n^2}{b_{d,q}} \quad (58)$$

where:

$$a_d = \frac{R_s}{L_{ds}}, a_q = \frac{R_s}{L_{qs}}, b_d = \frac{1}{L_{ds}}, b_q = \frac{1}{L_{qs}}$$

With a damping ratio of $\zeta=0.707$ and a d-q axis, the undamped natural frequency $\omega_{n,d,q}$ is based on the machine parameter (R_s, L_{ds}, L_{qs}) as in (3.34).

$$\omega_{n,d,q} = \frac{1}{1-\gamma} a_{d,q} \text{ with } 0 < \gamma < 1 \quad (59)$$

Finally, the inner loop PI gains are designed.

7.2.2 Speed control loop design

The speed PI-controller regulates the speed error to be zero during steady-state operation with a command torque signal as the control action, as shown in Fig. 27.

The command torque is entered into the plant speed transfer function. Equation (60) is the closed loop transfer function based on the controller gains.

$$CLTF = \frac{\left(\frac{Z_p}{J_m}\right)(K_p s + K_i)}{s^2 + \left(\frac{B}{J_m}\right)s} \quad (60)$$

Comparing the transfer function in (60) to the standard function in (57) gives the controller gains in (61).

$$K_p = \frac{2\zeta\omega_n J_m - B}{Z_p}, K_i = \frac{\omega_n^2 J_m}{Z_p} \quad (61)$$

With a damping ratio of $\zeta=1$ and a reduced undamped natural frequency of ω_n , the PI-controller gains are designed. Then, the gains should be tuned to further enhance the performance via MATLAB tools. The applied gains for the simulation are shown in Table 11 in appendix A. The command stator voltages are limited by the amplitude of the space vector voltage signal, which equals $1/\sqrt{3}$ times the DC-Link voltage for space vector pulse width modulation (SV-PWM). The d-q voltage limits are identified by the constraints shown in Fig. 28.

The d-q axis maximum command voltages [10] are presented in the following equations and illustrated in Fig. 29.

$$\sqrt{v_{ds}^2 + v_{qs}^2} = \frac{V_{dc}}{\sqrt{3}} \quad (62)$$

$$v_{qs_max}^* = \epsilon \frac{V_{dc}}{\sqrt{3}} \quad (63)$$

$$v_{ds_max}^* = \sqrt{1 - \epsilon^2} \frac{V_{dc}}{\sqrt{3}} \quad (64)$$

Fig. 27 Simplified speed control loop

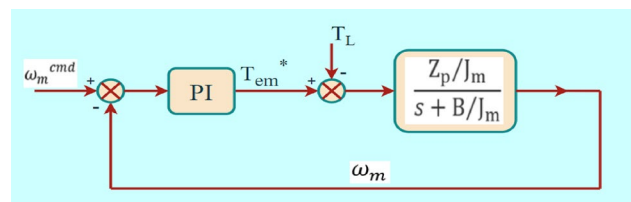


Table 11 The controller gains

Gain	Value
K_{pd}	332.8
K_{id}	1
K_{pq}	1489
K_{iq}	22.4
K_{pw}	3.047
K_{iw}	0.01

The command voltages of the d-q axis rotating reference frame are transformed to the three-phase reference frame, after which the SV-PWM algorithm is applied.

8 Simulation results and their discussion

The proposed FOC control technique incorporating both the MTPA and FW operating modes is modeled and simulated by MATLAB-Simulink for the IPMSM of Table 10. For the SV-PWM, a three-phase inverter model considering insulated gate bipolar transistors (IGBTs) is used, which is provided with a dc-link voltage of 540 V. The PWM frequency of the inverter is considered to be 4.050 kHz, and that for control is 10 kHz. The changeable speed characteristics of the IPMSM are discussed under fixed-load torque, as in case 1, and under sudden full-load torque, as in case 2.

8.1 Changeable motor speed under fixed load torque

The motor speed is changeable (1000–2500–3300 rpm). The disturbance load torque is constant at 4.3406 Nm. The simulation results are shown in Figs. 30, 31, 32. The results show that the command torque follows the maximum allowable torque as a function of the motor speed. The power trajectory is sustained at its maximum limit in the constant-power region. The IPMSM speed smoothly reaches the command speed during all motor operations. This voltage saturation

Fig. 28 Constraints for the d-q axes voltage and current

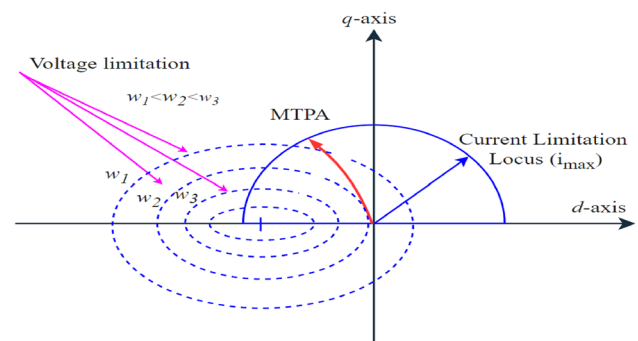
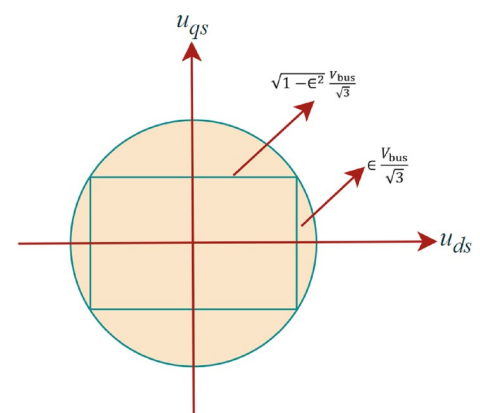


Fig. 29 d-q frame voltage constraints



causes a sudden magnitude reduction in torque and thus in the d-axis current. The d-q-axis FOC currents follow the command currents within the inverter limits precisely. The FOC currents and torque ripple during dynamic and steady-state operation, and the three-phase currents exhibit harmonic distortion, as shown in Fig. 31d. During high-speed operation with a rated power of 1500 W, FOC has a steady-state performance with a higher ripple in the torque and current waveforms. IPMSMs exhibit better performance in the low-speed range than in high-speed operation because of the high inductance of the permanent magnet flux.

8.2 Full motor load torque imposition at a fixed speed

Figures 33, 34, 35 show waveforms of the FOC under sudden full-load torque (7.162 Nm) at 3 s. The IPMSM is operated under the MTPA control region at a speed of 1000 rpm. The motor speed is constant and the same as the command, with only a 1% drop for the sudden load imposition. The torque and currents follow the commands precisely.

9 Discussion

The performance and significance of the presented results of the simulated Field-Oriented Control (FOC) technique, which incorporates Maximum Torque Per Ampere (MTPA) and Field Weakening (FW) operating modes has been discussed. This control approach is validated through MATLAB-Simulink simulations for an Interior Permanent Magnet Synchronous Motor (IPMSM) model, examining key responses under both variable-speed operation with fixed load torque and sudden full-load torque conditions.

9.1 Relevance and added value

The proposed FOC technique exhibits high precision in managing IPMSM speed and torque under diverse conditions, making it suitable for applications that demand a wide dynamic speed range and efficient motor control. This method maintains maximum power within the constant-power region and enables smooth speed and torque transitions, reflecting its effectiveness in both low- and high-speed operations. Our results emphasize the ability of this approach to address torque precision and speed control challenges, even under abrupt load variations. In comparison to traditional FOC strategies, the combination of MTPA and FW modes in our control technique improves motor efficiency and responsiveness, which is particularly valuable for applications like electric vehicles and industrial drives.

9.2 Comparison with existing works

The findings align with prior research employing MTPA and FW techniques for optimized IPMSM performance. For instance, Schoonhoven Garin, (2016), [90] demonstrated a robust nonlinear speed control method based on MTPA and

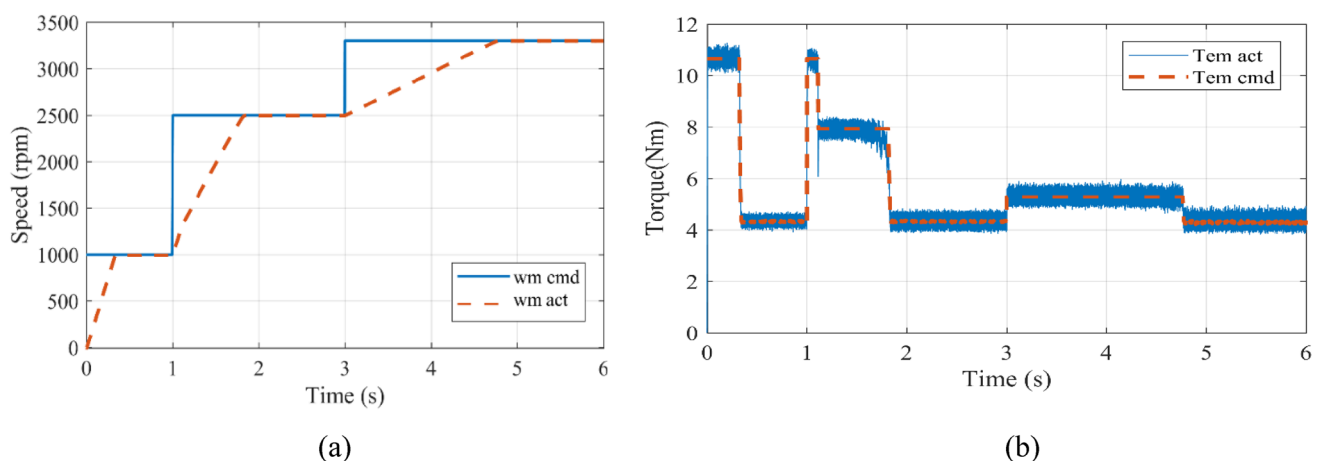


Fig. 30 Rotor speed and electromagnetic torque under constant load **a** motor speed, and **b** torque

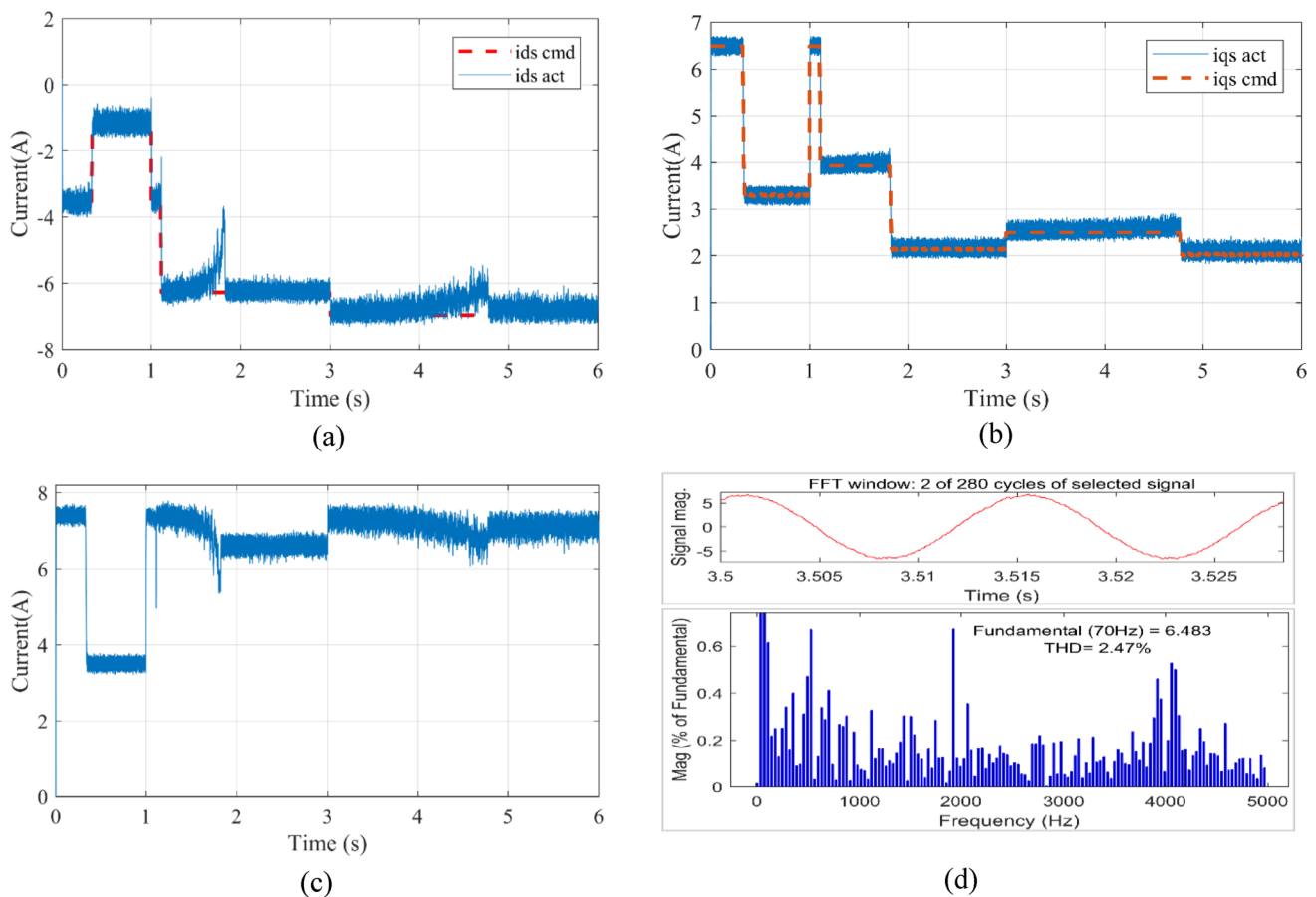
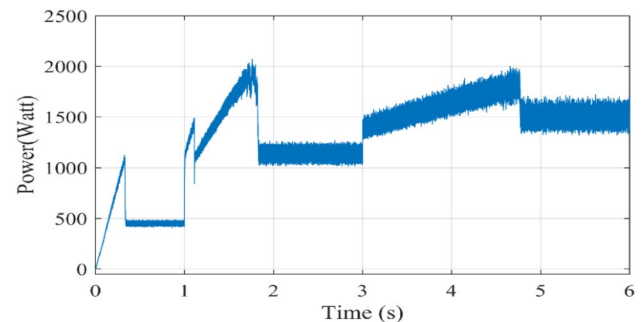


Fig. 31 Motor currents and THD of the three-phase current **a** i_{ds} , **b** i_{qs} , **c** i_{ds} , **d** i_{abcTHD}

Fig. 32 Motor power



FW for a wide-speed-range IPMSM, utilizing Lyapunov stability criteria and an adaptive backstepping technique. Their approach effectively manages nonlinearities and ensures robust system stability, with both experimental and simulation results confirming excellent drive performance, particularly in extended speed ranges and load disturbance rejection. Similarly, Qureshi and Sharma (2024) [91] presented a novel approach integrating MTPA and MTPV for enhanced wide-speed-range control, focusing on flux-weakening techniques that limit current and reduce losses, resulting in improved motor efficiency and performance. Their work also highlights the benefits of MTPA for higher speed ranges and MTPV for the full speed range.

While both studies provide strong control strategies, our approach offers additional insights by integrating the MTPA and FW schemes for precise speed and torque control across a broader range of operating conditions. In contrast to the methods discussed [90, 91], which are more focused on torque efficiency and stability, our technique extends the operational capability of the IPMSM, ensuring smooth transitions and maintaining power within the constant-power region, even under varying load conditions. Furthermore, the presented simulation results explicitly demonstrate transient and

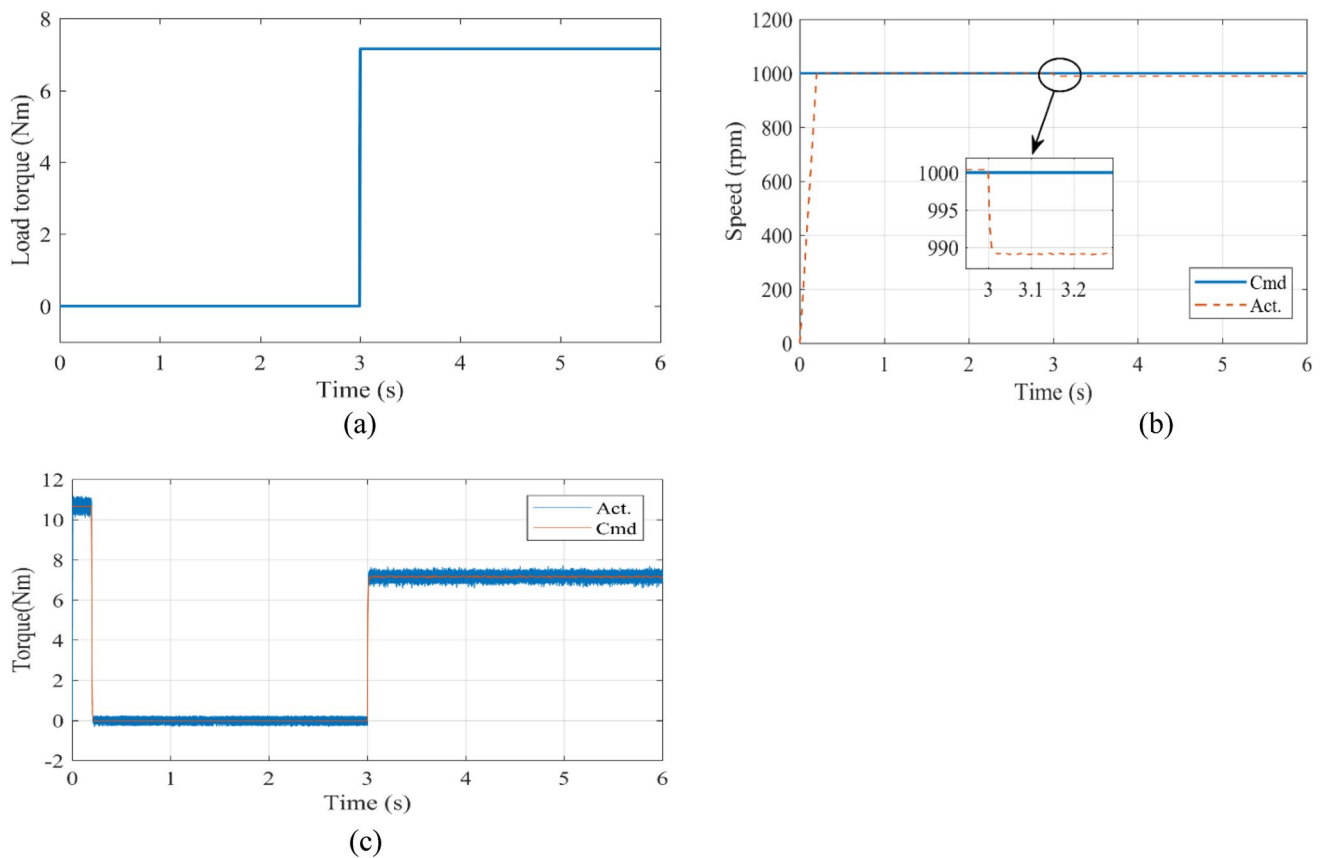


Fig. 33 Rotor speed and electromagnetic torque under constant load (a) motor load (b) motor speed and (c) torque

steady-state responses, showcasing rotor speed, electromagnetic torque, and motor currents under various operating scenarios, which were not as extensively addressed in the prior works.

9.3 Future work and limitations

Future work will aim to provide more analysis considering the sensitivity of permeameters uncertainty. Additionally, further efforts to minimize torque ripple at high speeds will enhance the robustness and applicability of the proposed technique.

10 Conclusions

This paper presents a comprehensive review of two drives based on field-oriented control (FOC) and finite control set-model predictive control (FCS-MPC) techniques for IPMSM. Moreover, this paper proposes and implements an IPMSM drive system based on an FOC with a conventional PI controller under different operating modes. The FOC scheme for an IPMSM drive has been proposed and implemented under the MTPA/FW mode, fixed-load torque, and sudden full-load torque imposition. The MTPA/FW was implemented within the voltage source inverter limits of the current and voltage. The proposed method can be used at 5–100% rated speed region, and it also has good performance in the flux weakening region. The simulation results showed that FOC has good performance over MTPA/FW dynamic and steady-state operations.

This paper presents an inclusive approach to the model, control, and operation of an IPMSM to be appropriate for engineers and researchers in motor drives. Future work should be conducted to improve the performance of reduced ripple, another proper control technique such as MPC, and a sensorless observer for performance applications such as

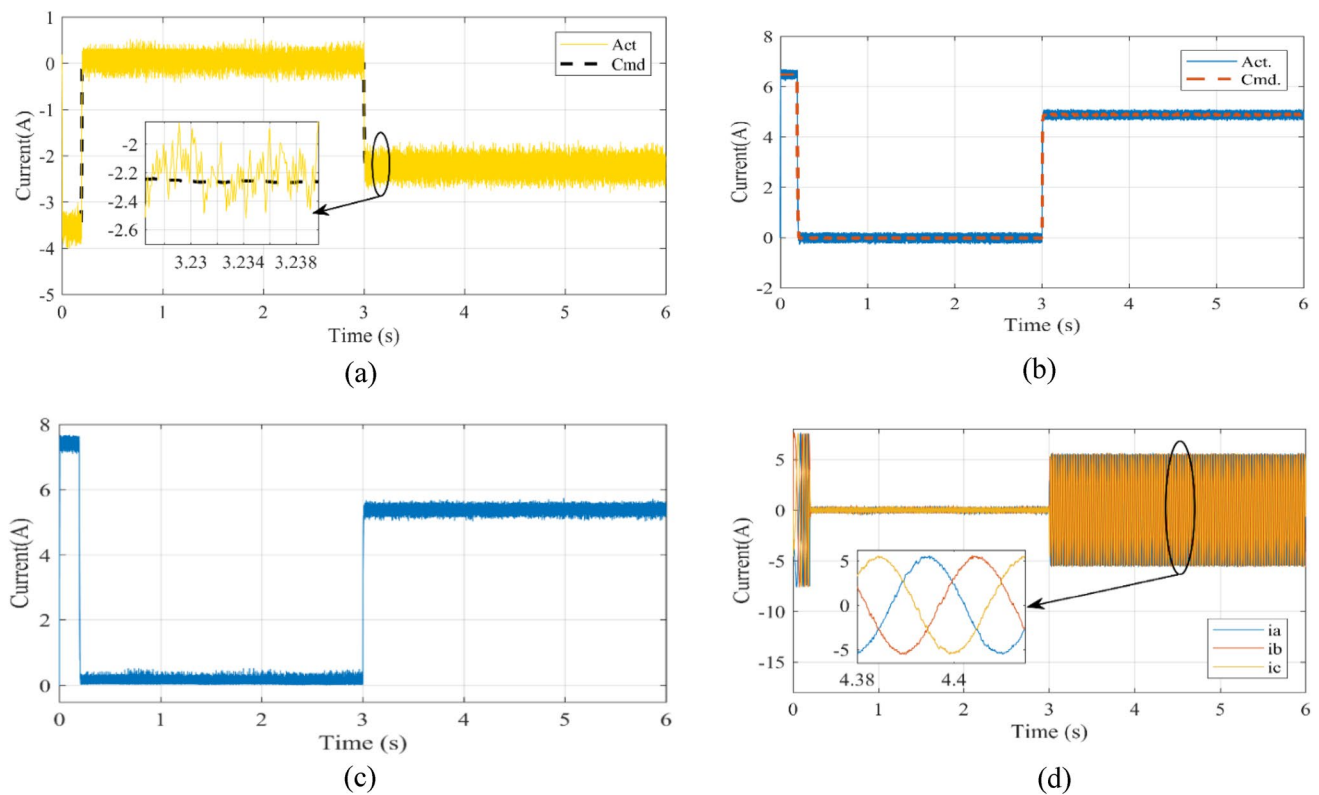


Fig. 34 Motor currents and THD of the three-phase current **a** i_{dsr} , **b** i_{qs} , **c** i_{sr} , **d** i_{abc}

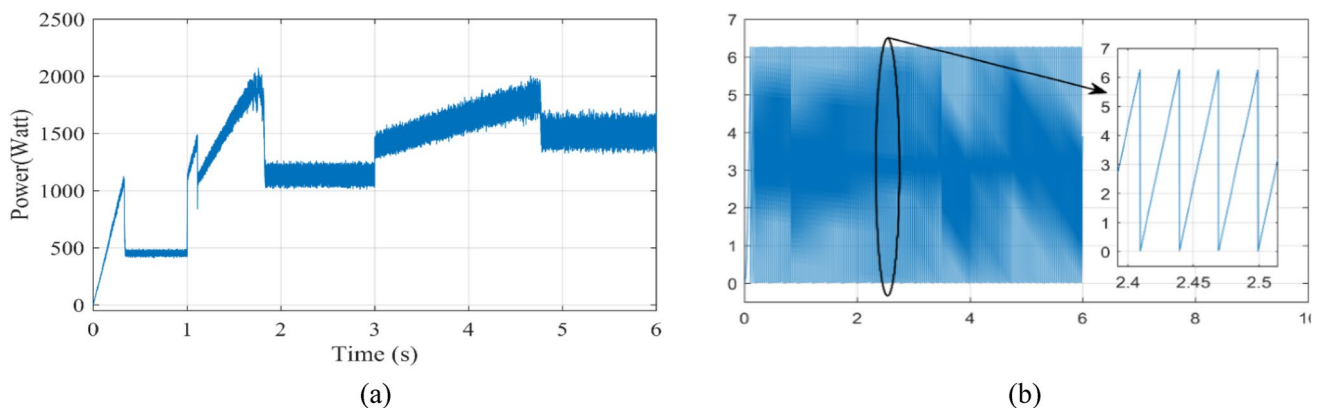


Fig. 35 Motor power and rotor position **a** power, and **b** Rotor position in rad

traction applications. Moreover, parameter estimation issues may be employed in future work to enhance the IPMSM performance and predictive maintenance.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interest The authors declare no competing interests.

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Appendix A

The space vector is a commonly applied tool for AC machine drives to represent the vectors of flux, voltage, and current magnitudes in a compact manner. Models and differential equations of the machines and inverters are derived by using the space vector representation [33]. Model analysis based on space vector representation is easier than that based on three-phase representation. After introducing the space vector concept, the models of the inverter and the IPMSM will be discussed. A balanced three-phase two-pole machine is assumed, as shown in Fig. 36.

Applying a three-phase balanced supply on the stator windings produces three-phase sinusoidal stator fluxes in the ideal case. The three-phase sinusoidal fluxes a , b , and c at the axis of angle θ with respect to the phase a axis are:

$$\varphi_a(\theta) = \varphi_m \cos(\omega t) \cos(\theta) \quad (65)$$

$$\varphi_b = \varphi_m \cos(\omega t - 120) \cos(\theta - 120) \quad (66)$$

$$\varphi_c = \varphi_m \cos(\omega t - 240) \cos(\theta - 240) \quad (67)$$

The resultant stator air gap flux at the axis of angle θ is:

$$\varphi_r(\theta) = \frac{3}{2} \varphi_m \cos(\omega t - \theta) = \frac{3}{2} \varphi_m \angle -\theta \quad (68)$$

Now, the three-phase machine is converted to a two-axis fixed frame (α - β) machine to be easy to control. To maintain the same magnitude as the three-phase variables, we multiplied the vector by $2/3$ [32]. Generally, for any variable x , we obtain the space vector shown in Fig. 37.

$$x = \frac{2}{3} (x_a + ax_b + a^2 x_c) \quad (69)$$

where:

$$a = e^{j(2\pi/3)} \quad (70)$$

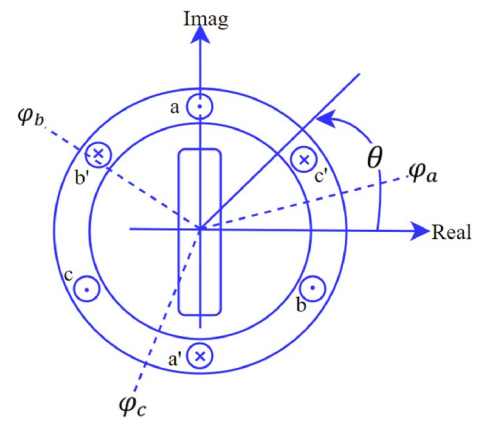
$$x = x_\alpha + jx_\beta \quad (71)$$

The balanced three-phase components are transformed to the α - β coordination as in Eq. (73).

$$\begin{bmatrix} x_{\alpha s} \\ x_{\beta s} \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -0.5 & -0.5 \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix} \quad (72)$$

The transformation matrix T is called the Clark direct transformation, where:

Fig. 36 Two-pole three-phase PM. Machine configuration



$$T = \frac{2}{3} \begin{bmatrix} 1 & -0.5 & -0.5 \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \quad (73)$$

The magnitudes of the α - β coordination are equal to the three phase magnitudes and still vary sinusoidally, shifting by 90° . Thus:

$$|x_{as}| = |x_{\beta s}| = |x_{as}| = |x_{bs}| = |x_{cs}| \quad (74)$$

The direct transformation operation is represented in Fig. 38.

The inverse transformation from $\alpha\beta$ coordination to the three phases is performed by the inverse matrix T:

$$\begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1/2 & \frac{\sqrt{3}}{2} \\ -1/2 & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} i_{\alpha s} \\ i_{\beta s} \end{bmatrix} \quad (75)$$

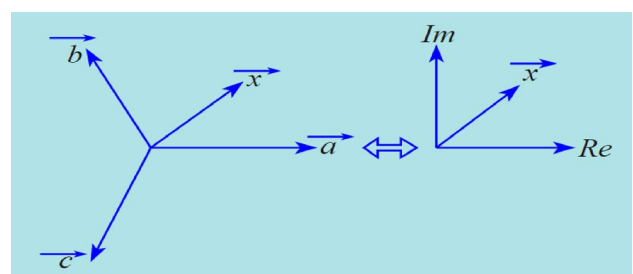
The inverse Clark transformation matrix relation is represented in Fig. 39.

θ represents the angular rotor position that is calculated from the angular speed as follows:

$$\theta = \int \omega dt = \omega t \quad (76)$$

Representing the space vector in the synchronously rotating reference frame is highly useful for developing the dynamic model of the IPMSM. The stationary reference frame α - β is transformed to the synchronously rotating reference frame (d-q) based on the synchronous angular speed. The rotating speed of the d-q reference frame is the synchronous speed so that the space vector to the d-q frame does not rotate. The d-q frame space vector is represented by constant real and imaginary magnitudes. For all machines except for the synchronous frame, there is another rotating reference frame (D-Q) that is fixed on the rotor structure. Because of the rotor speed in the synchronous machine, the synchronous speed results in the D-Q and d-q frames are the same. The space vector representation in different rotating frames is shown in Fig. 40.

Fig. 37 Space vector representation in the $\alpha\beta$ reference frame



The d-q frame variables are held constant, which makes using the frame in control easy. The transformation from fixed coordination to the rotating coordination frame is performed by using the rotational transformation. To transform from the $\alpha\beta$ to the d-q frame, Eq. (77) is used.

$$\begin{bmatrix} x_{ds} \\ x_{qs} \end{bmatrix} = \begin{bmatrix} \cos\theta_m & -\sin\theta_m \\ \sin\theta_m & \cos\theta_m \end{bmatrix} \begin{bmatrix} x_{\alpha s} \\ x_{\beta s} \end{bmatrix} \quad (77)$$

where θ_m is the rotor position calculated in Eq. (78).

$$\theta_m = \int \omega_m dt = \omega_m t \quad (78)$$

The direct rotational transformation matrix M is:

$$M = \begin{bmatrix} \cos\theta_m & -\sin\theta_m \\ \sin\theta_m & \cos\theta_m \end{bmatrix} \quad (79)$$

The direct rotational transformation takes another form in Eq. (80) and is shown in Fig. 41.

$$\vec{x}^r = e^{-j\theta_m} \vec{x}^s \quad (80)$$

The transformation from d-q coordination to the stationary frame is performed by using the inverse rotational transformation in Eq. (81).

$$\begin{bmatrix} x_{\alpha s} \\ x_{\beta s} \end{bmatrix} = \begin{bmatrix} \cos\theta_m & -\sin\theta_m \\ \sin\theta_m & \cos\theta_m \end{bmatrix} \begin{bmatrix} x_{ds} \\ x_{qs} \end{bmatrix} \quad (81)$$

where:

Fig. 38 Representation of the direct transformation matrix relation

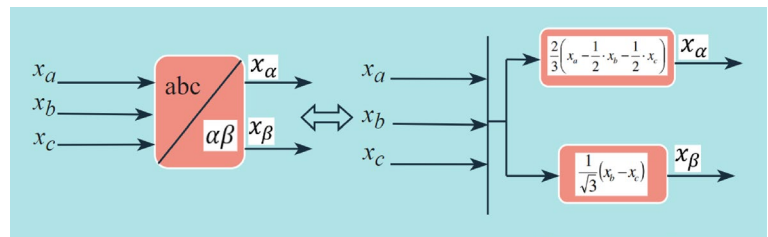


Fig. 39 The inverse Clark transformation matrix representation

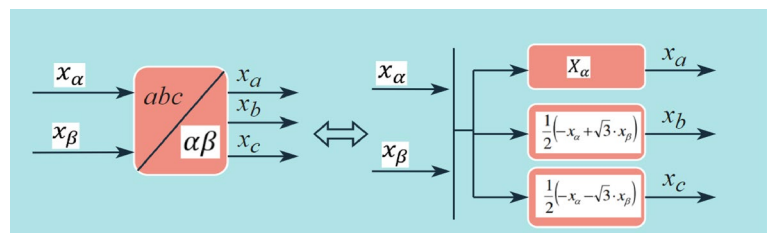


Fig. 40 The space vectors are represented in various frames

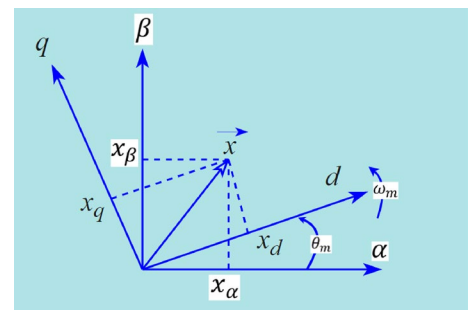
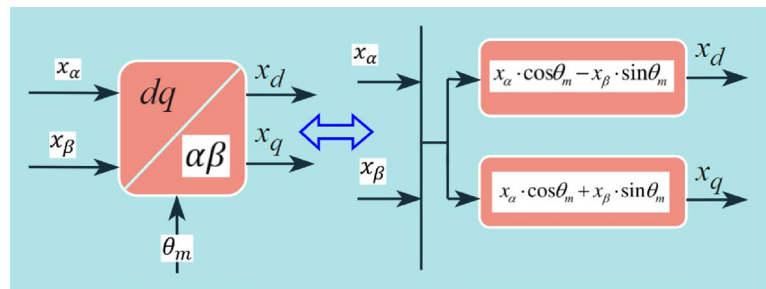


Fig. 41 Direct rotational transformation representation



$$M^{-1} = \begin{bmatrix} \cos\theta_m & -\sin\theta_m \\ \sin\theta_m & \cos\theta_m \end{bmatrix} \quad (82)$$

There are several methods for observing the mechanical parameters of drive systems, which can be classified into three main categories [92–95]:

1. *Periodic Perturbations*: This technique applies periodic changes to the electromagnetic torque or rotor speed, resulting in a forward-reverse transient torque associated with the moment of inertia [92, 93].
2. *Startup or Step Response Calculation*: In this approach, the moment of inertia is calculated during motor startup or step response. The resulting transient torque is non-zero, remaining constant for steady acceleration or deceleration, and varying for fluctuating rates [94].
3. *Loaded Condition Estimation*: This method estimates parameters under loaded conditions, treating the load as a constant disturbance torque. Recursive Least Squares (RLS) and similar techniques are used to estimate time-varying inertia and load torque, useful in applications like low-speed induction motor speed monitoring [95].

Moreover, the damping coefficient can be experimentally determined by measuring the system's response to a known input, such as observing the motor's deceleration rate when power is cut off. This "free response" method involves recording the rate at which the motor's speed decays due to internal frictional forces, from which the damping coefficient can be calculated by analyzing the deceleration curve.

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